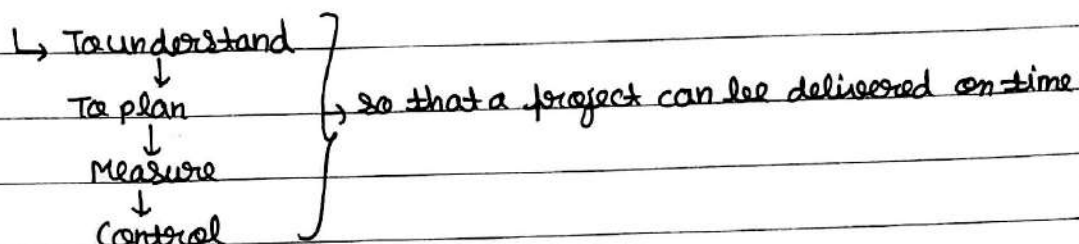


Software Project Management (SPM)

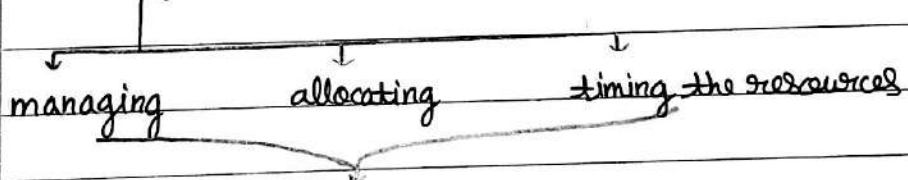
Software Project Management (SPM) is discipline of achieving targets in software while optimising the use of resources i.e. time, people, money, energy over the course of project.

Goal of SPM:-



Detailed description of SPM:-

1) Process of



to develop a software that meet requirements.

2) SPM is sub-discipline of project management in which software projects are:

- a) Planned.
- b) Implemented.
- c) monitored.
- d) controlled.

Framework:-

Step 1) Problem Identification.

Step 2) Problem Definition. (Proposal creation)

Step 3) Project Planning.

Step 4) Project Organization.

Step 5) Resource Allocation.

Step 6) Project Scheduling.

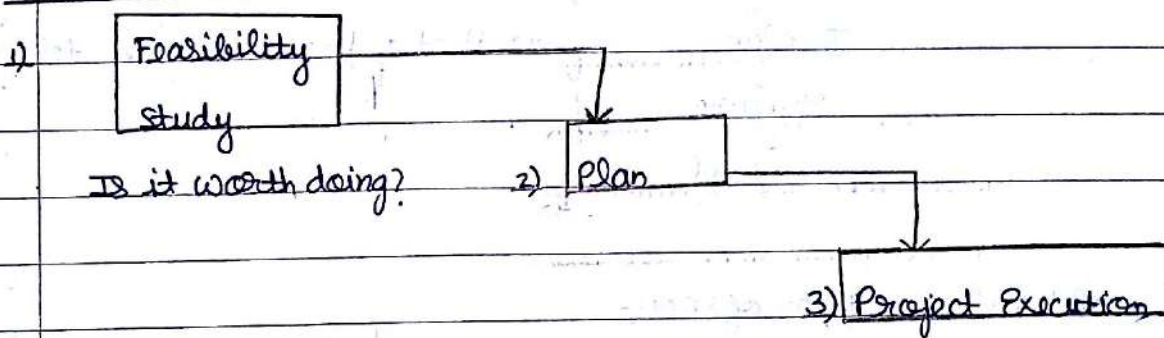
Step 7) Tracking/Reporting.

Step 8) Project Termination.

Example: Project

Manager Developer Analyst

Activities in SPM:-



Software Development life cycle (SDLC):-

Graphical/textual representation of software that how it works.

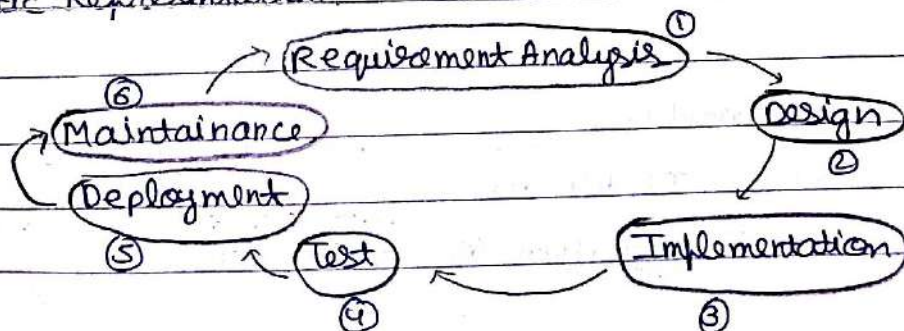
* How to start a project.

* How to manage

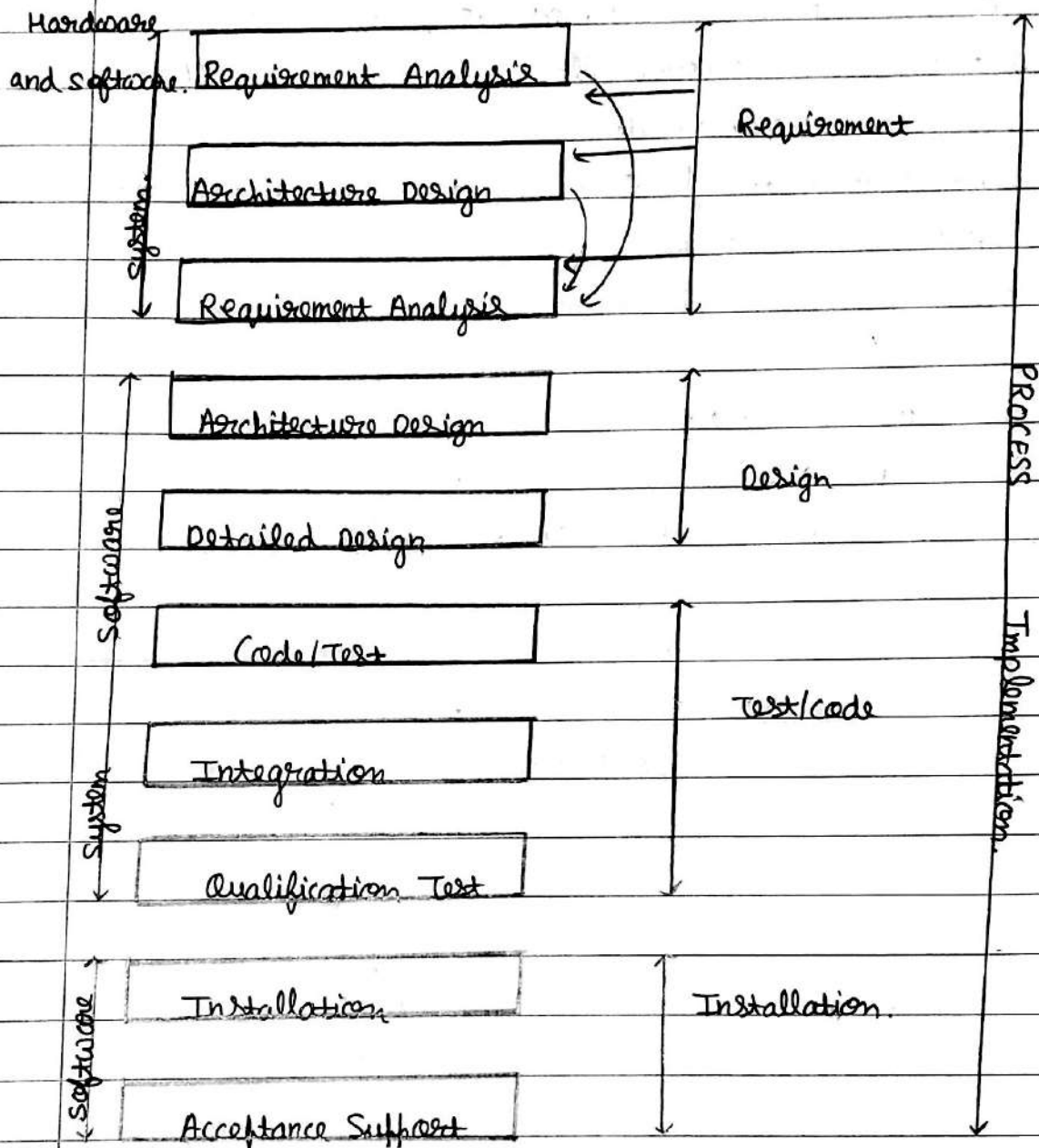
* How to replace/alter.

A life cycle model of a software is a graphical or textual representation of its life. Process model may describe details of various types of activities carried out during different phases or document produced.

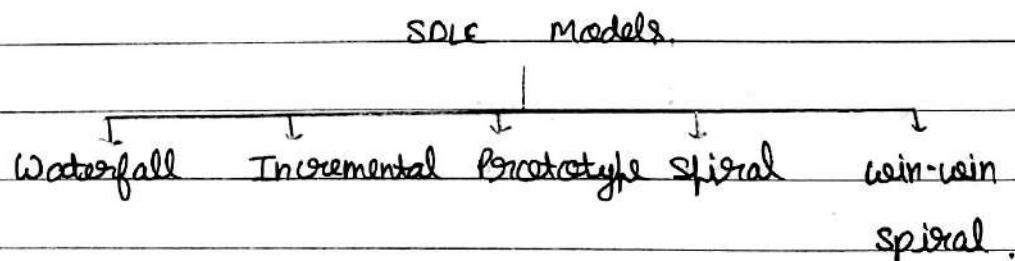
Diagrammatic Representation:-



Detailed Diagrammatic Representation:-



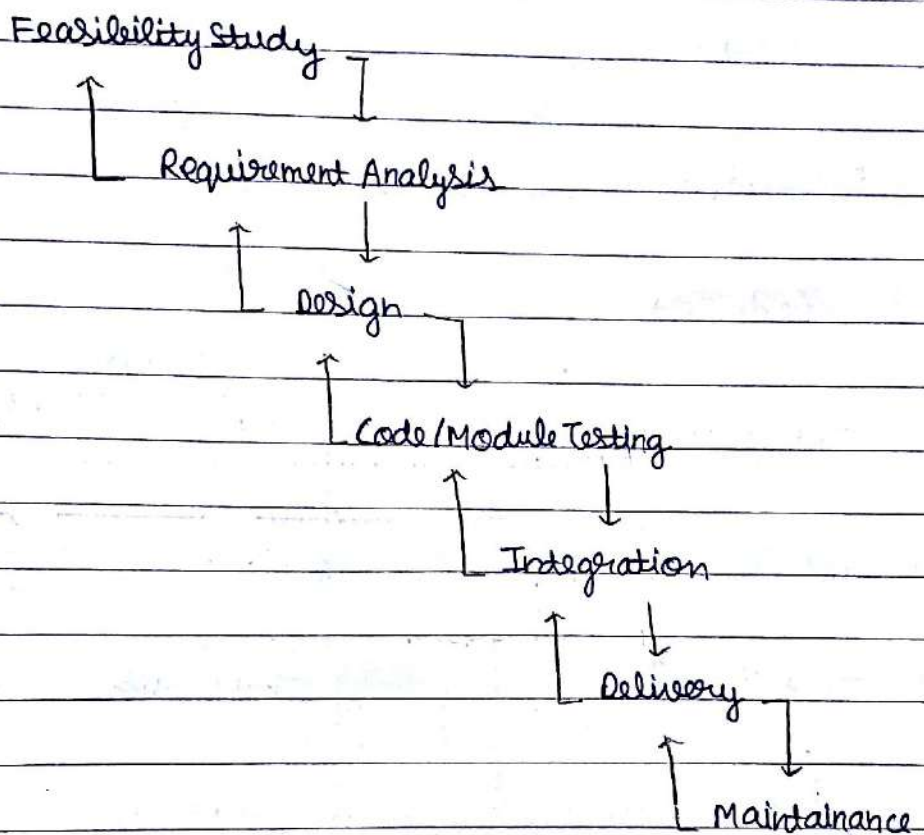
SDLC Models:-



Waterfall Model:-

- 1) This is the earliest approach used in Software Development.
- 2) It is in linear sequential flow. Hence, it is also referred as linear sequential life cycle model.
- 3) Any phase in the development process begins only when the previous phase is complete.

* Diagrammatic Representation:-



* Applications:-

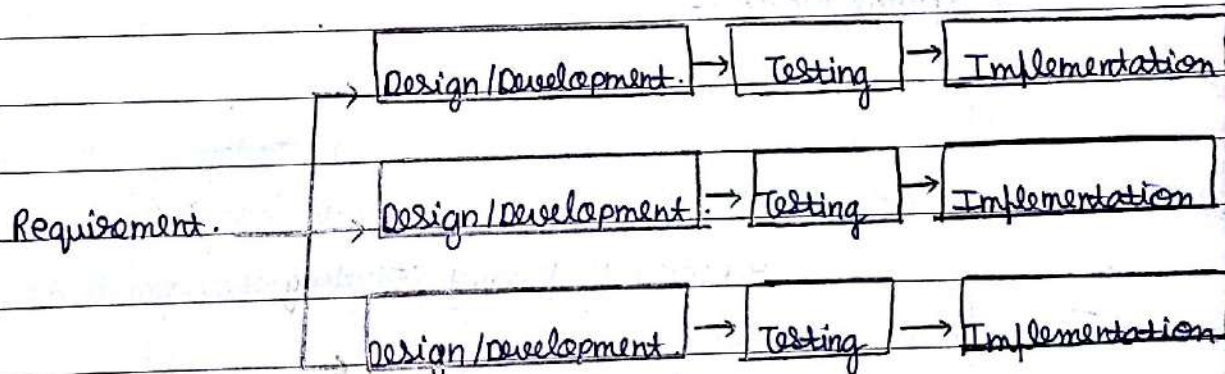
- 1) Product definition is stable.
- 2) Requirements are very well documented, clear and fixed.
- 3) Technology is understood and not dynamic.
- 4) Project is short.

* Advantages:-

- 1) Simple and easy to use.
- 2) Easy to manage.
- 3) Clearly defines stages.
- 4) Easy to arrange tasks.
- 5) Process and results are well documented.

* Disadvantages:-

- 1) High amount of risk.
- 2) Not suitable for complex models.
- 3) Poor model for long projects.
- 4) Cannot accommodate changing requirements.

Incremental / Iterative Model:-

Basic idea behind this model is to develop a system through repeated cycles and in a smaller portion of time.

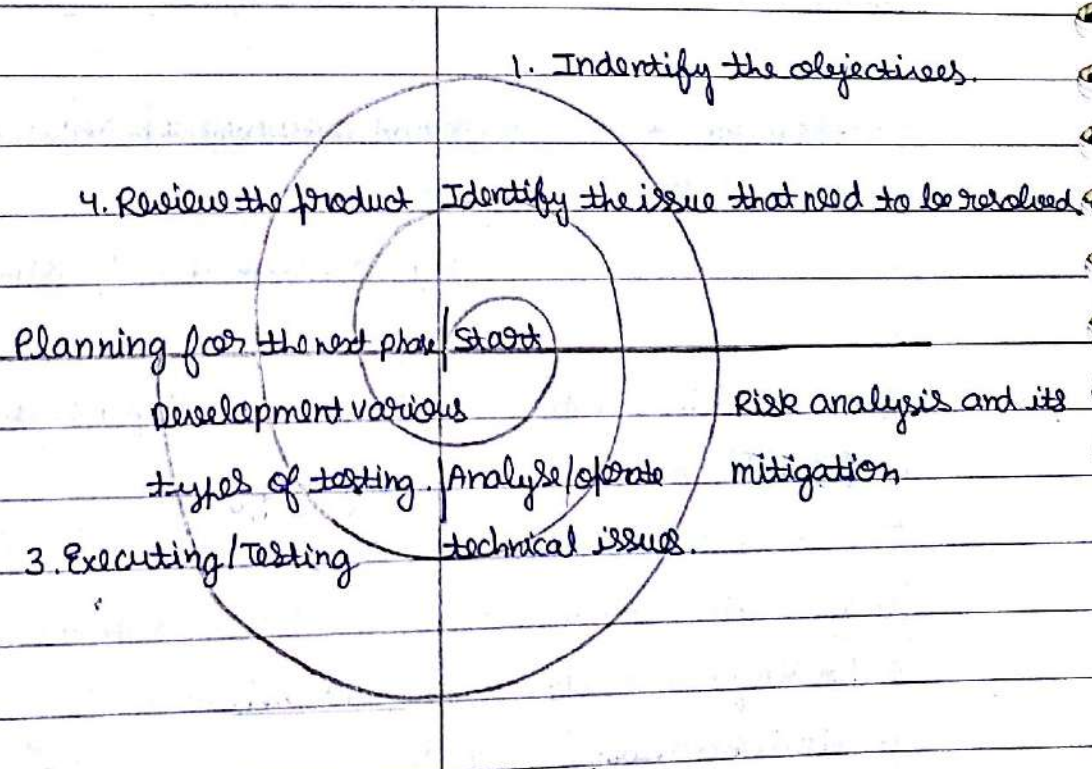
In this model, whole requirements is divided into various builds. During each iteration, the development module goes through design, implementation and testing phases. Each subsequent release of the model at function to previous reviews. The key to successive use of an iterative development life cycle is rigorous validation of requirement, verification, testing of each version of software.

* Advantages:-

- 1) Results obtained easily and periodically.
- 2) Better suited for large and complex projects.
- 3) It supports changing requirement.
- 4) Easier to manage.
- 5) Operating time is less.

* Disadvantages:-

- 1) Not suitable for small projects.
- 2) More management attention is required.
- 3) More resources may be required.
- 4) Highly skilled resources are required for risk analysis.
- 5) Project progress is dependent on risk analysis phase.

* Spiral Model:-

1) It allows incremental release of a product or incremental refinement through each iteration around the spiral.

2) Higher emphasis is given on risk analysis.

* Applications:-

1) It is widely used in software industry as it is synchronised with the natural development process of any product i.e. learning with maturity and also involves minimum risk for the customers.

2) For medium to high risk projects.

3) Used for long term project commitment.

* Advantages:-

1) Better risk management.

2) Requirements can be captured more accurately.

3) Changing requirements can be accommodated.

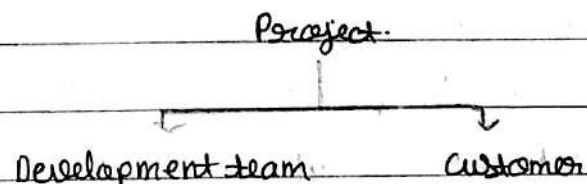
* Disadvantages:-

1) High cost.

2) Management is more complex.

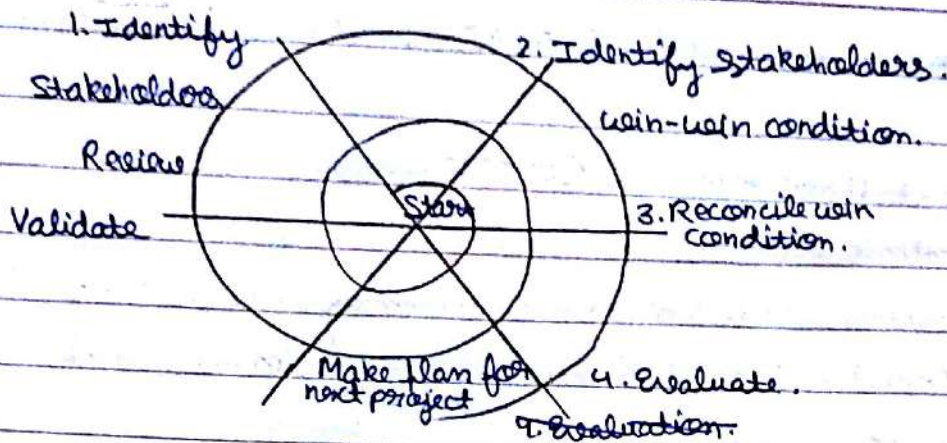
3) Large no. of intermediate stages requires excessive documentation that makes the process time consuming.

* Win-Win Spiral Model:-



(Stake holders → developers or owners).

(Diagram at next page).



It is an extension of spiral approach. The only difference is that at the time of identifying the requirements, the development team and customers hold the discussion and negotiate ^{on} the requirements that need to be included in current iteration of software. The approach is called win-win because it is a winning situation for both developer and customer.

* Customer wins:-

By getting the product that fulfills most of the requirements.

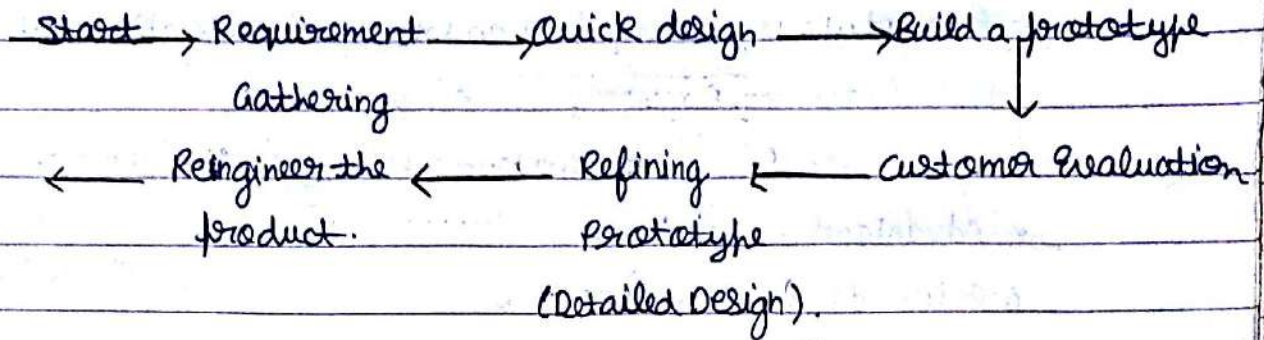
* Development team wins:-

* By delivering a software which is developed with all the requirements established after negotiating with the customer. This approach is generally used in "TIME-BOUND" release.

* Prototype Model:-

Instead of freezing the requirements, before a design of coding ~~and~~ can proceed, a prototype is built to understand the requirements.

(Diagram is at next page).



The goal is to provide the system with overall functionalities.

* Advantages:-

- 1) Users are actively involved in development.
- 2) Errors can be detected much earlier.
- 3) Missing functionalities can be identified easily.
- 4) Better understanding of systems being developed.

* RAD (Rapid Application Development) Model:-

* Main aims of RAD Model:-

- 1) To decrease the time taken and cost incurred to develop software systems.
- 2) To limit the cost of accommodating change request by incorporating them as early as possible before large investment being made of developing and testing.

* Explanation:-

Cost of accommodation change was extremely high in other models as they take long time and enormous cost to have a good solution. The RAD model tries to overcome this problem by incorporating customer feedback on successfully developed prototypes. Development takes place in a series of short cycles called iteration (cycle). Plans are made for one iteration at a time only. Time planned for each iteration is called time box. During each cycle, a quick and rough prototype for some functionalities developed. The customer evaluates the prototype and gives feedback.

Prototype is nothing but an instrument to gather customer feedback

Not released x ← **PROTOTYPE**

↳ "Instrument to gather customer feedback"

* Advantages:-

- 1) Reduce development time.
- 2) Increase reusability of component.
- 3) Quick reviews may occur.
- 4) Encourage customer feedback.

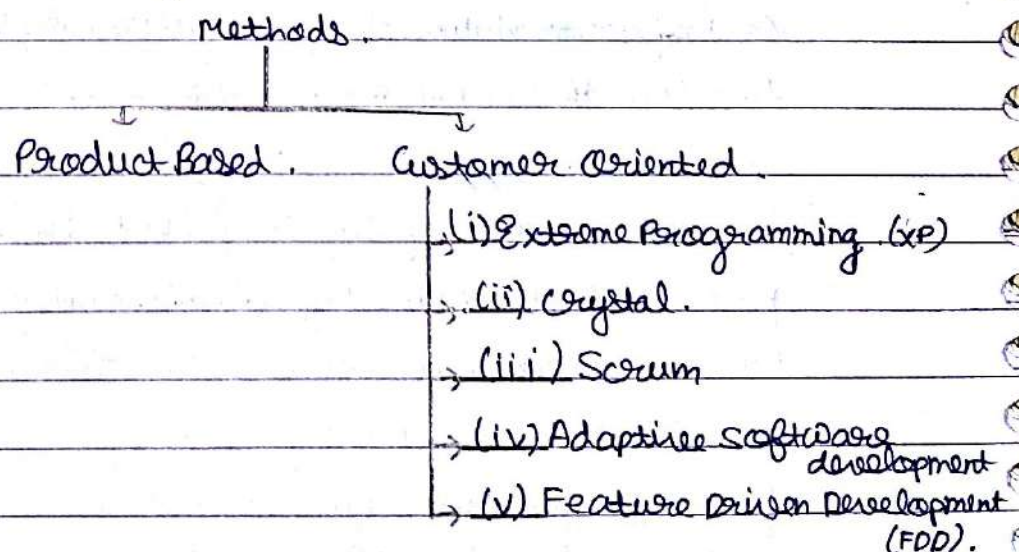
* Disadvantages:-

- 1) Requires highly skilled developer.
- 2) High dependency on modelling.

* Agile Methods:-

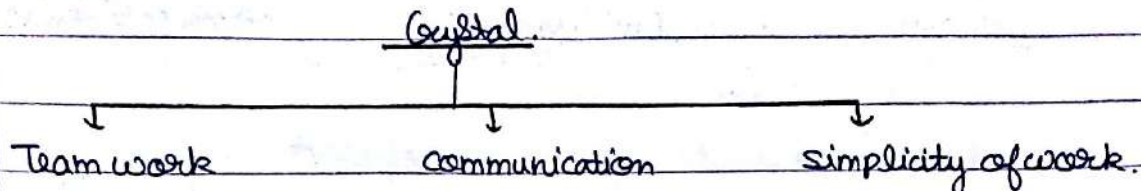
It is a software development approach that is based on incremental/iterative approach where the solutions and requirements evolve through the collaboration between the organization and its team members.

- 1) Requirement specifications.
- 2) Series of versions.
- 3) System and user interface.



(i) Extreme Programming: This is one of the type of methodologies of agile methods which enhance the quality of the software and responsive to the requirement.

(ii) Crystal: It is one of the most light-weight and adaptable approach to software development.



(iii) Scrum: It emphasize on customer feedback, team = self management, striving to build properly, tested product implement with short cycles.

(iv) Adaptille Software development: It is based on principles such as:-

- a) Business needs.
- b) Active user involvement.
- c) Frequent delivery.
- d) Integrated testing.
- e) Stakeholder collaboration.

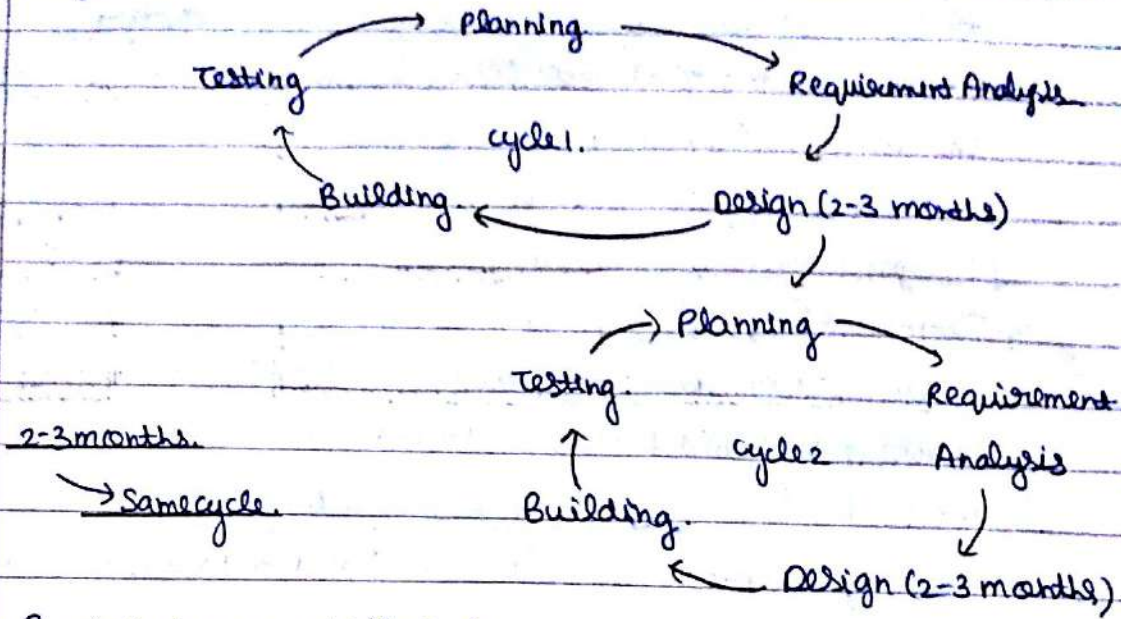
(v) Feature Driven Development (FDD): It is a model driven a short iteration process begins with establishing an overall model shape.

• Features of Agile Programming:-

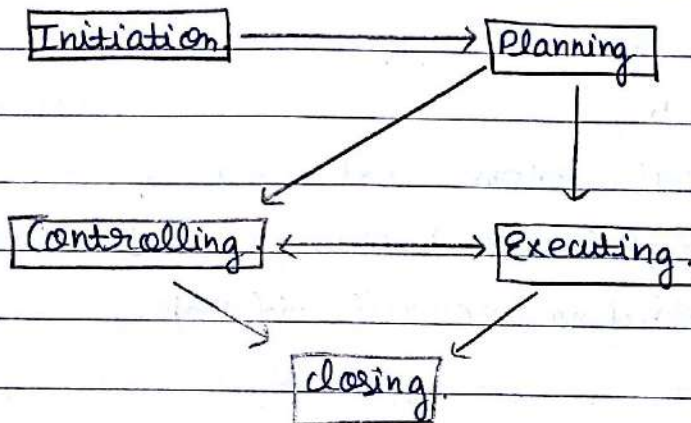
- 1) It is simple and easy to use.
- 2) Customer involvement.
- 3) Increment delivery.
- 4) People not process (People or human being are more important).
- 5) Embrace change.

* Diagram at next page.

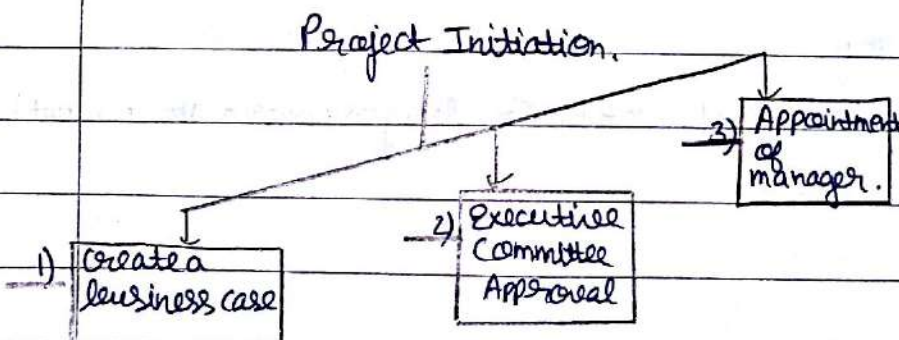
• Diagram:-



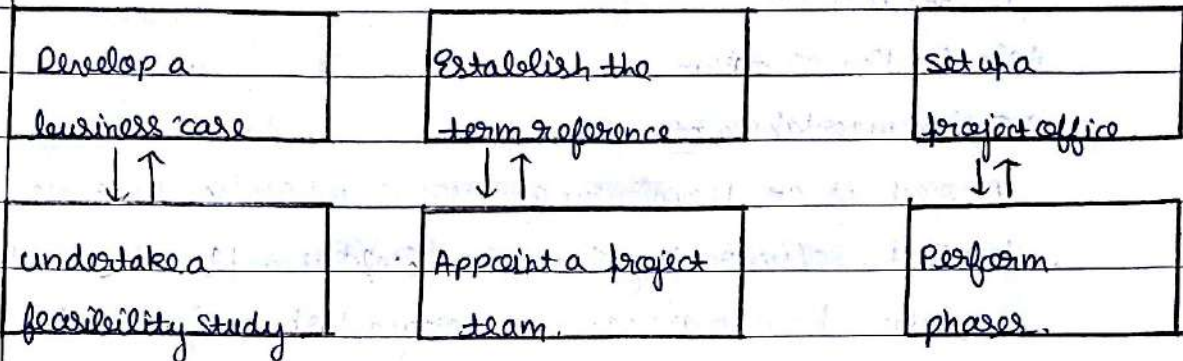
• Project Management life cycle:-



1) Project Initiation:-



It involves starting up the project by documenting a business case, term reference, appointing the team and setting up the project office.



* Advantages:-

- 1) This model is the most practical model in nature.
- 2) Promotes team work.
- 3) Functionality can be developed rapidly and demonstrated.
- 4) Resource requirements are minimum.
- 5) Good model for environment that changes steadily.

* Disadvantages:-

- 1) Not suitable for handling very complex dependencies.
- 2) More risk of sustainability.
- 3) Strict delivery management.
- 4) Depends heavily on customer's interaction.

PROJECT:-

→ Time, cost, resource, specifically.

Project is a:-

1) Specific plan or design.

2) A plan undertaking or

Project is a temporary endeavour undertaken to create a unique product. Software project can be defined as a planned activity that describes how we are going to carry a task before start.

• Characteristic features of project:-

1) Planned.

2) Aiming at specific target.

3) Carried out for customer.

4) Carried out by temporary work group.

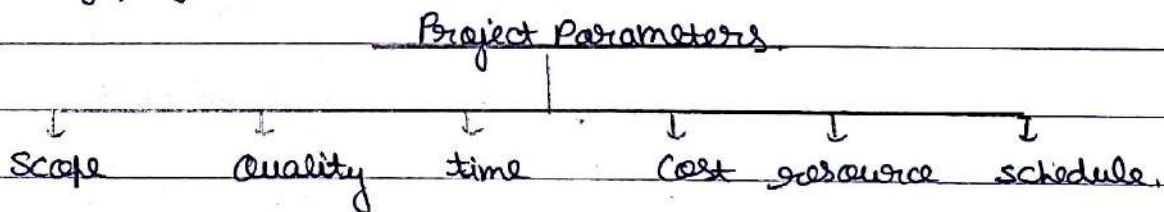
5) Consists of different phases.

6) Constraint by time, cost and resource.

7) Large or complex.

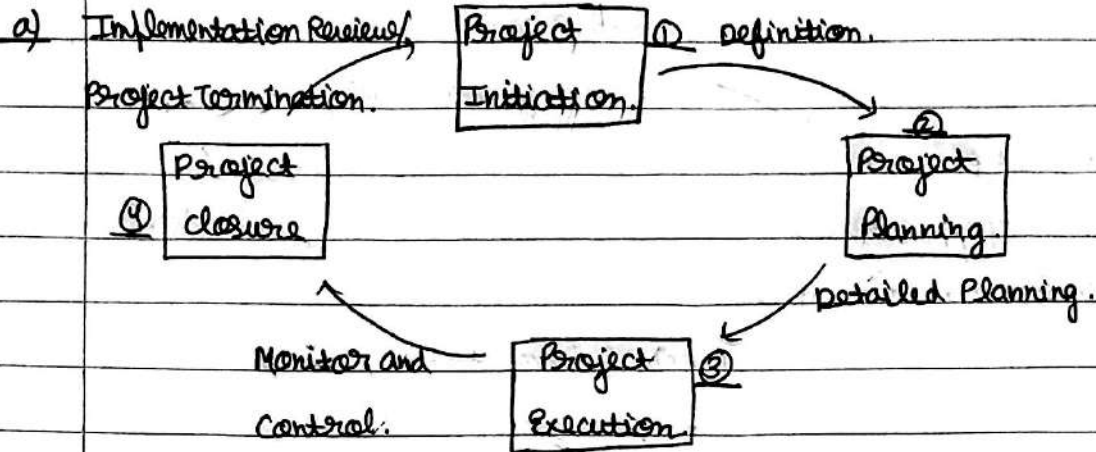
• Project Parameters:-

Parameters are those specific points that affect the the project or factors affecting project.



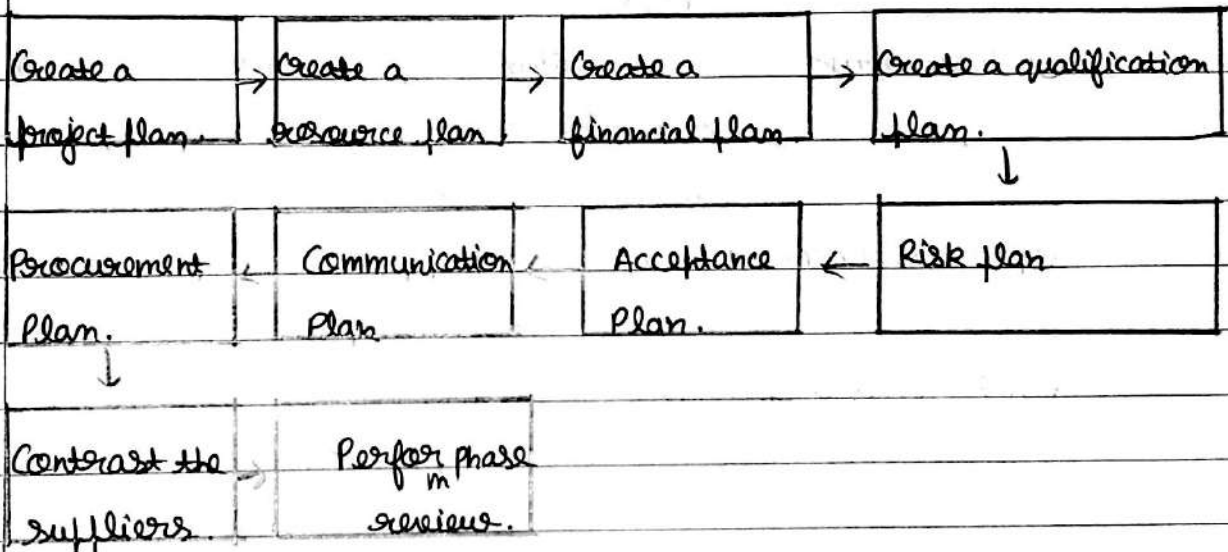
• Principle of project management:-

(next page).



b) Project Planning :-

- Project Initiation Document.
- Communication Plan.
- Executive Committee Approval.



Software Project Management.

Unit II: Project Planning.

Project Planning (Introduction):-

Developing a realistic plan is essential to gain an understanding of the resources required that how these should be applied.

Types of Project Plan.

- 1) Quality plan. 2) Validation plan. 3) Configuration management plan. 4) Staff Development plan. 5) Maintenance plan.

① Quality plan: It describes quality procedures and set standards for it.

② Validation plan: Describes standards that ~~will~~ be used in validating or testing a project.

③ Configuration Management Plan: Describes installation procedures and ~~also~~ state of work products that are under configuration control.

④ Staff Development plan: Describes how the skills and experience of the project team members will be developed.

⑤ Maintenance plan: Predicts the maintenance requirements of the system and its cost.

Activities of Project plan:-

① Feasibility study → Feasibility report

② Requirement Analysis → User requirement

③ Prototype Development → Evaluation Report

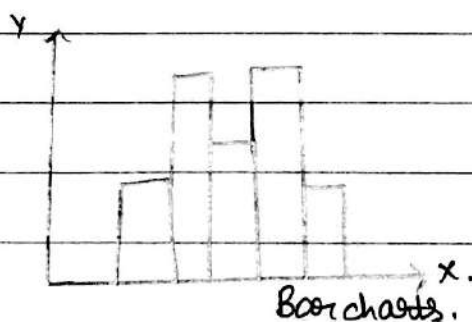
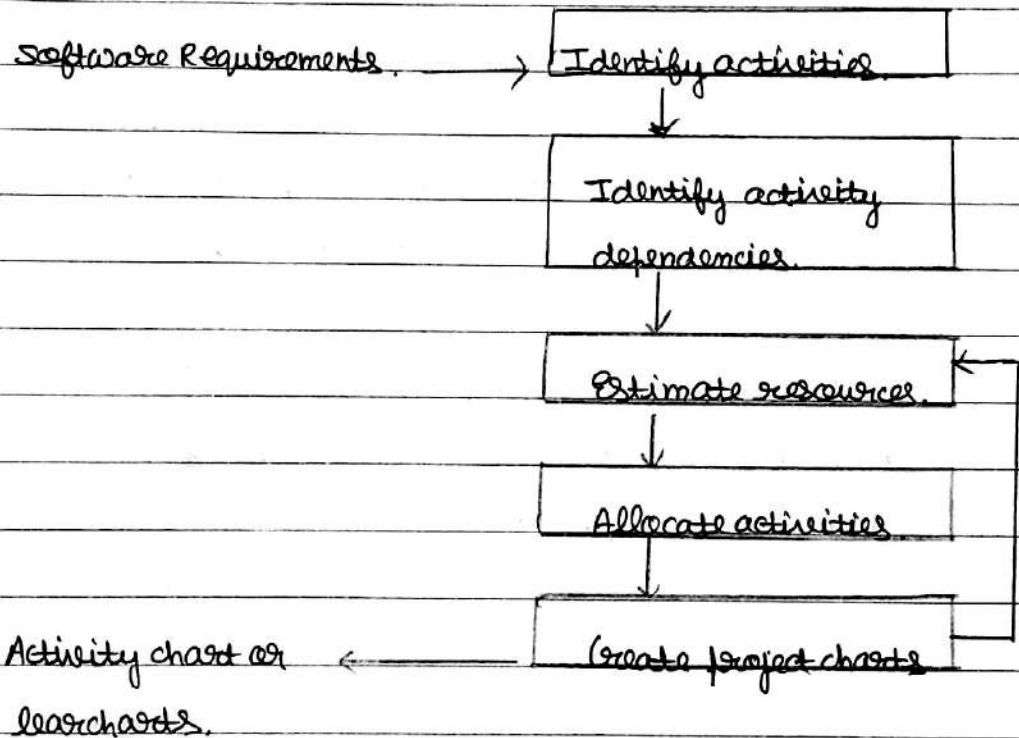
④ Design study —————> Architectural Design.

⑤ Requirement specification —————> System Requirements.

• Project scheduling:-

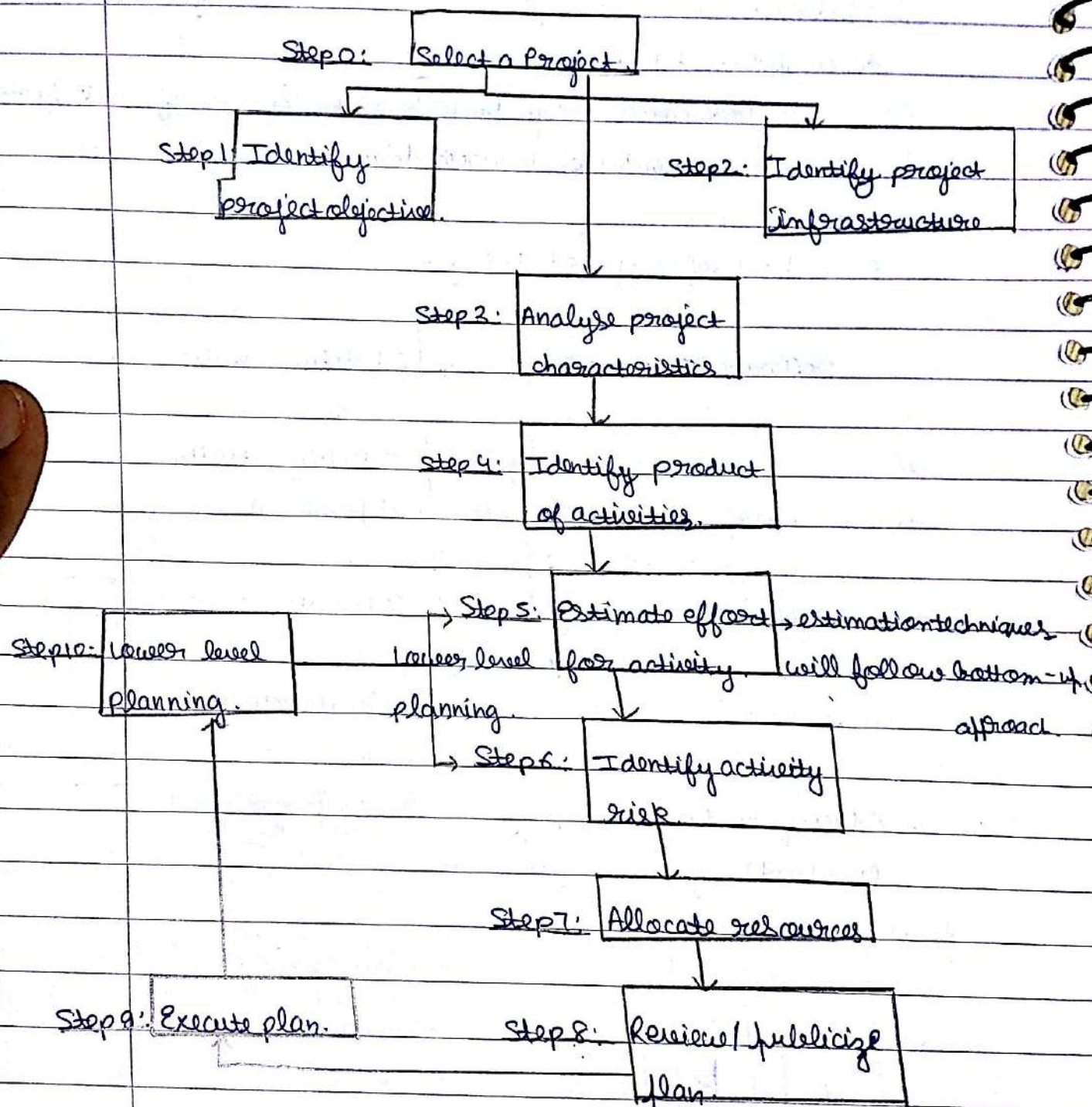
- ① Split tasks into no. of tasks and estimate time and resources require to complete each task.
- ② Organize task concurrently to make optimum use of work force.
- ③ Minimize task dependencies to avoid delay.

• Flow chart of Project scheduling:-



Scheduling explores possible ways of allocating limited resources across task. Project scheduling involves separating the total work involved in a project into separate activities.

Activities in project planning:-



① Step 1: Identify project objective:-

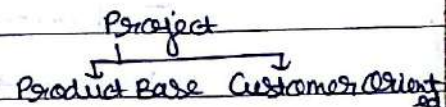
- Identify project authority.
- Identify stakeholders.
- Establish methods of communication between all parties.
- Identify

② Step 2: Identify project infrastructure:-

- Establish relationship between project and planning.
- Identify team organization.
- Build prototype.

③ Step 3: Analyse project characteristics:-

- Distinguish the project as product driven or customer driven.



④ Step 4: Identify product of activities:-

- Document generic product and recognize product instability.

⑤ Step 5: Estimate effort for activity:-

- ~~Effort~~ Estimation techniques will be followed that will follow bottom-up approach.

⑥ Step 6: Identify activity risk:-

- Identify and quantify activity based risk.
- Plan risk reduction.

⑦ Step 8: Review/publish plan:-

- Review quality aspects of project plan.
- Document plan and obtain agreement.

② Step 3: Execute Plan:-

a) This step may require reiteration of planning process.

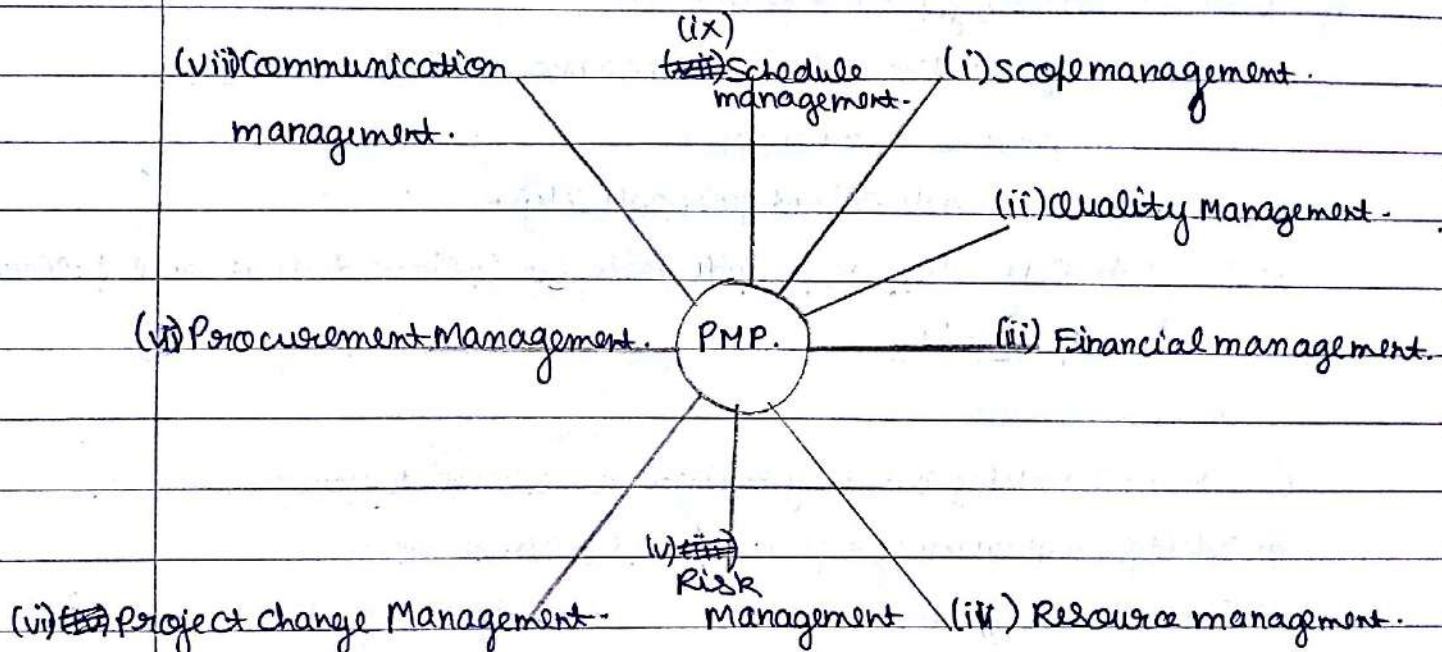
• Project Management Plan (PMP)

PMP is a formal approved document that defines how the project is executed, monitored and controlled. It may be summary or detail and may be composed of one or more subsidiary management plans and other planning document.

Objective of PMP is to define the approach to be used by project team to deliver the intended project management scope of the project.

The project manager creates a plan following input from the project team and team stakeholders.

• Main aspects of Project Management Plan (PMP):-

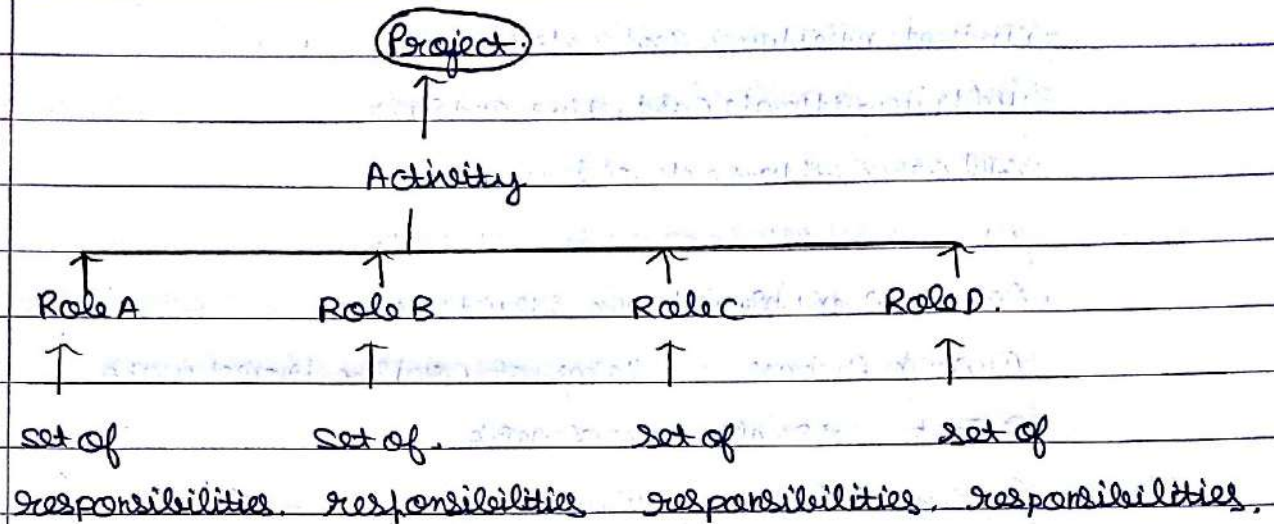


Procurement Management:-

When to do, what to do, how to do? to start a project.

• Activity Resource Requirement:-

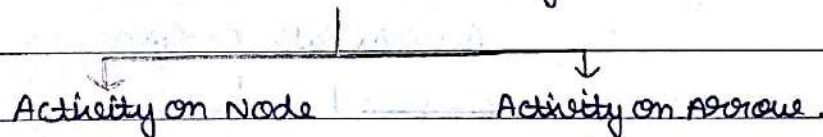
It identifies the type and quantities of resource needed for each activity in a work package including assumptions that were made in determining the type of quality.



• Network planning:-

It is a type of scheduling technique model that projects activities and their relationships as a network.

Network Planning



① Activity on Arrow:-

This technique is used to visualise the project as a network whose activities are drawn on arrows, joining circles or nodes.

② Activity on Node:-

where activities are represented on node and links between nodes represent precedence requirement.

• Work Breakdown Structure (WBS):-

Dividing complex and large projects into simpler and manageable tasks is the process identified as work breakdown structure.

* ~~Reasons~~ Reasons for creating WBS:-

- 1) Accurate and ~~breakable~~ project readable project organization.
- 2) Indicate milestones and control points.
- 3) Helps to estimate cost, time and risk.
- 4) Illustrate the scope of project.

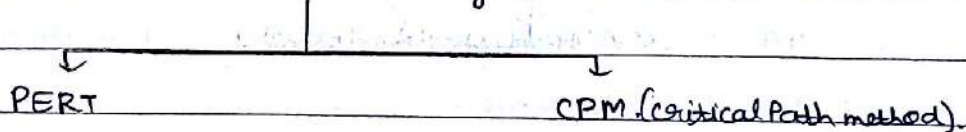
• Activities of Breakdown Structure.

- ① Work Package. → large or complex project work.
- ② Task. → smaller piece of work.
- ③ Deliverables. → output.
- ④ milestones. → final target.

• ~~Work Package~~ Critical Path Analysis:-

- 1) Critical path is the sequence of activities that takes longest time to complete.
- 2) Any delay to an activity in the critical path causes delay to overall project.

Critical Path Analysis.



① PERT (Project Evaluation Review Technique).

- Optimistic time.
- Pessimistic time.
- Most likely time.

a) Most likely time: Time we would expect to complete a task under normal circumstances. This is denoted by 'm'.

- b) optimistic time: shortest time in which we could expect to complete the activity. It is denoted by 'a'.
- c) Pessimistic time: worst possible time allocated for all reasonable eventualities. It is denoted by 'b'.
- * Expected duration = $\frac{(a + 4m + b)}{6}$

• Difference between PERT and CPM:-

<u>PERT</u>	<u>CPM</u>
1) It is a project management technique or tool which is suitable for projects that have unpredictable activities.	1) CPM is used for projects having predictable activities.
2) PERT uses three estimates for time.	2) CPM use single estimate for time.
3) PERT is a probabilistic project management tool.	3) CPM is a deterministic project management tool.
4) PERT does not allow project management planner to determine which aspect of the project to sacrifice when a trade off is needed ^{in order to} to complete the project.	4) ^{CPM} PERT allows for the same.

1st Mid Term:-

Starting from 15th Feb, 2016.

Course: I unit and II unit. #.

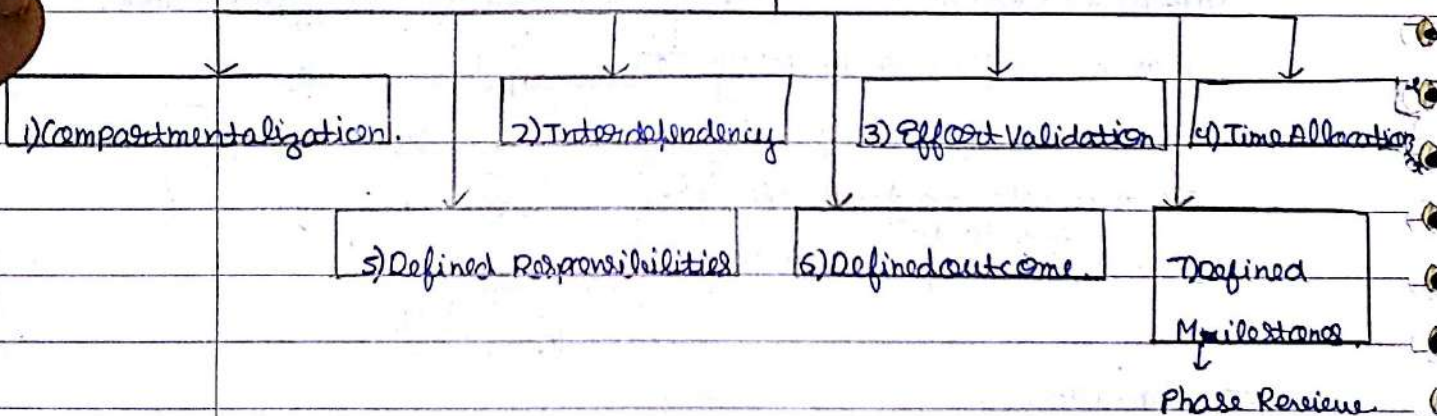
Software Project Management-unit III.

Project Scheduling: Monitoring and control.

- why projects get delayed?
 - underestimating the resources.
 - unrealistic deadlines established out of the team.
 - changing customer requirements.
 - ~~under unrealistic risk~~ that were not considered.
 - miscommunication among project staff.
 - Failure by project management team to recognise project falling under which schedule.

Project Scheduling Principles:-

Project Scheduling Principles:-



- ① Compartmentalization: In this step, product and process must be decomposed into no. of activities.
- ② Interdependency: Task that can be completed parallelly must be separated from those that can be completed serially.
- ③ Effort Validation: Tasks are distributed to the staff members to complete them within the time estimated using effort estimation techniques.

What makes you happy?

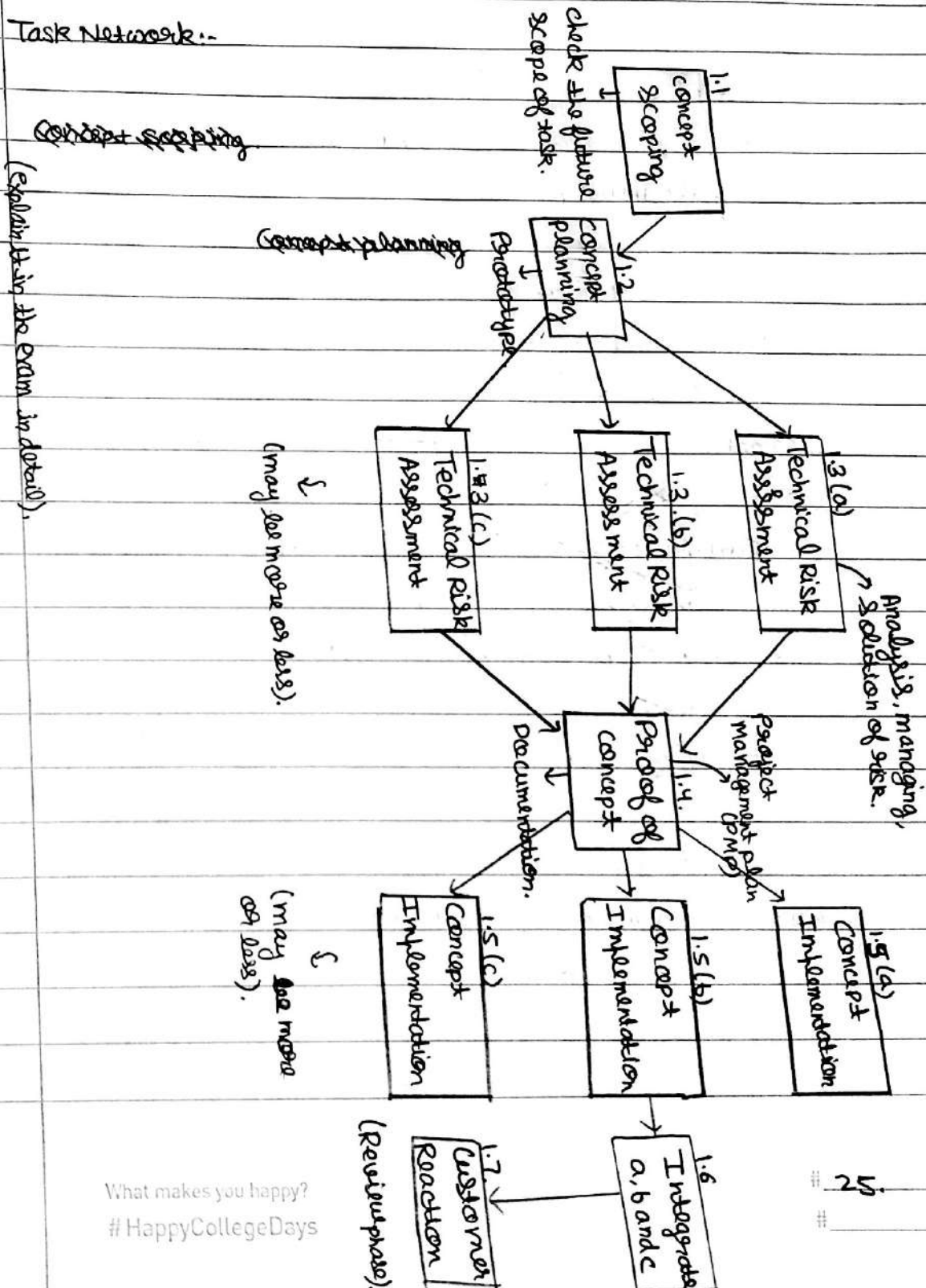
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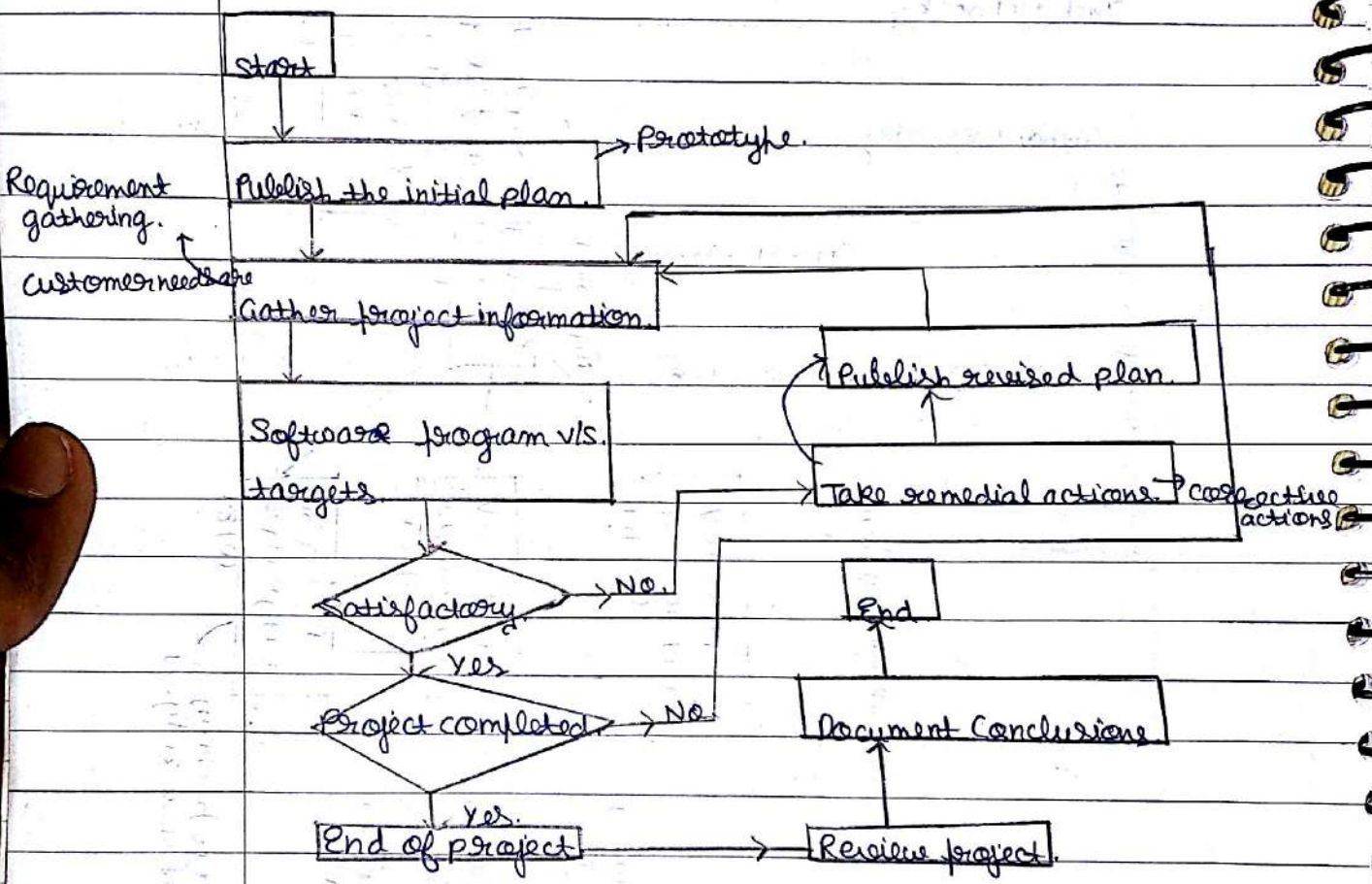
- ④ Time Allocation: Assigning a task to starting and ending dates.
- ⑤ Defined Responsibilities: Assigning a particular task to a particular team member.
- ⑥ Defined Outcome: Every task must have defined output.
- ⑦ Defined milestones: A milestone is accomplished to pass a quality review check.

Task Network:-



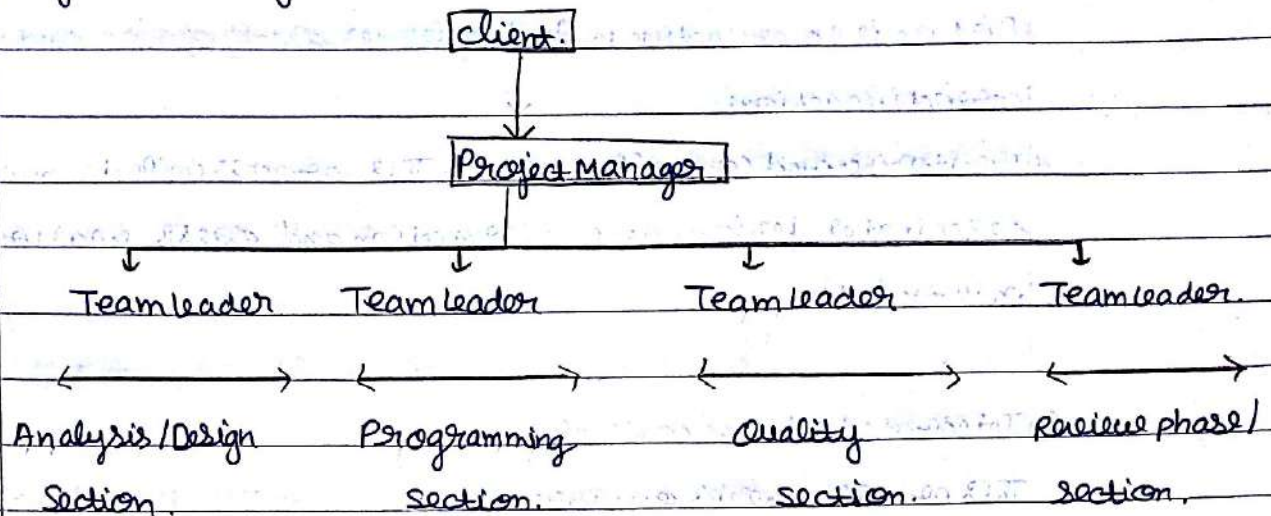
- Monitoring and Controlling:-
 Supervise. Risk Management.

① Project Control cycle: It shows how once the initial project plan has been published, project control cycle is the continual process cycle of monitoring progress. Against the plan where necessary revision of the plan has been taken into account.



Progress $\longleftrightarrow$ Targets
 what have much we have achieved. what to achieve.

② Project Reporting Structure:-



• Monitoring and Controlling Processes:-

① Monitoring and Controlling Work.

② Integrated change control

③ Scope verification.

④ Scope control

⑤ Schedule control

⑥ Cost Control

⑦ Reperforming quality control

⑧ Managing the project team.

⑨ Performance reporting

⑩ Managing stakeholders.

① Monitoring and Controlling Work:-

While project being executed, the project is monitored, controlled by implementing levels of corrective actions. Project is measured to ensure that whether it is within the acceptable variance of time, cost and scope.

- * Output (Monitoring and Controlling):-
- Recommended corrective and preventive actions.
 - Recommended defect repairs.
 - Request changes.
 - ~~Scope verification~~

Date

17 02 16.

Stakeholders → developers/owners!

* Purpose:-

Purpose is to be proactive in finding issues ahead of time and taking corrective actions.

* Monitoring and Controlling Work: This process collects, measures and disseminates performance information and assess measures to forecast potential items.

② Integrated change control:-

This process works on corrective actions taken throughout the process.

* Output:-

- Approved change request.
- Rejected change request.
- Updates the project management plan.
- Approved corrective and preventive actions.
- Approved defect repairs.
- Deliverable.

③ Scope Verification:-

This ensures that the project deliverables are accepted.

* Output:-

- Accepted deliverables/outputs.
- Recommended corrective actions.

④

④ Scope control:-

Scope of the project is controlled.

* Output:-

- updates to project scope statement and scope base line.

What makes you happy?

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- b) update to work Breakdown structure.
- c) Requested changes.
- d) updates to organizational change assets.
- e) updates to project management plan.

⑤ Schedule Control:-

This process monitors the project schedule.

* Output:-

- a) updates to schedule model data.
- b) Performance measurements.

⑥ Cost Control:-

It controls cost and changes to project budget.

* output:-

- a) cost estimate updates.
- b) Cost baseline updates.
- c) Performance measurements.

⑦ Performing quality control:-

This process is used to determine whether the project is meeting quality standards or not.

* output:-

- a) quality control measurement.
- b) Validate defect repair.
- c) Updates to quality baseline.

⑧ Managing the project team:-

Tracks team performance and provides feedback to resolved issues.

* Output:-

- Request changes.
- Recommended corrective and preventive changes.

⑨ Performance Reporting:-

This process collects and distributes performance information.

Performance Reporting:-

- ↳ Progress Report.
- ↳ Status Report

* Output:-

- Performance reports are generated.

⑩ Managing Stakeholders:-

Manage communication among the developers and team.

* Output:-

- Resolved issues.
- Approved change requests.
- Approved corrective actions.
- Updates to project management plan.

• Software Configuration Management (SCM):-

SCM is concerned with tracking and controlling changes to the software.

Why SCM?

* Problem:-

- Multiple people have to work on a software i.e. changing.

6) More than one version of the software has to be supported.

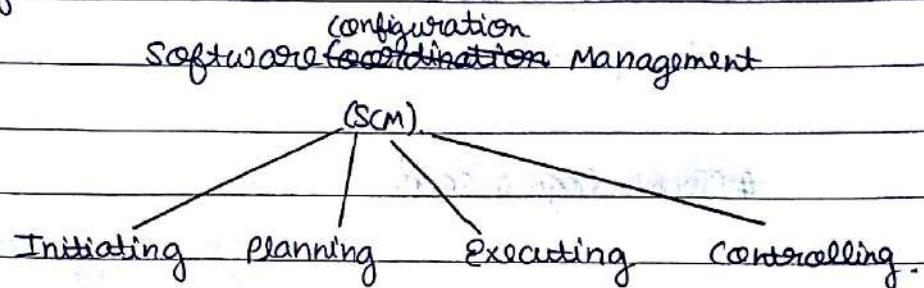
* Need of Coordination.

* Manages evolving software systems.

Definition:-

SCM is a set of management disciplines within the software engineering process to develop a baseline.

SCM (Software ^{Configuration} ~~Coordination~~ ~~Management~~) encompasses the techniques of:-



Purpose of SCM:-

1) Problem associated with concurrent access:-

~~It is an~~ Important reason for configuration management is to control the access to the different deliverables of objects.

2) Undoing changes:-

It becomes ~~to~~ easy to undo some part of a revision to a certain versions.

3) System accounting:-

It denotes keeping track of those who made a particular change to a configuration item, what change exactly made and when.

knowing
 → who
 → what
 → when.

4) Handling variants:-

Keep track of all variants i.e. variable and their versions and revision.

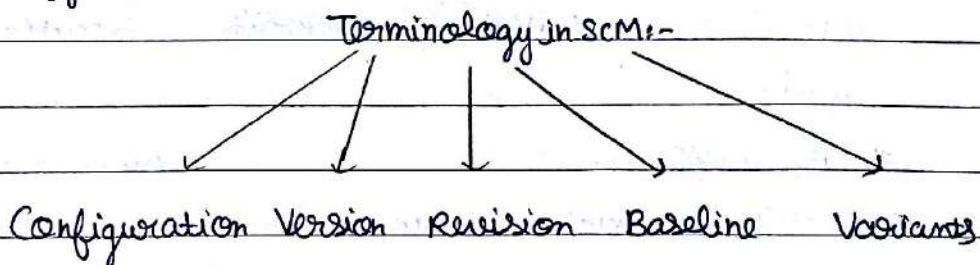
5) Accurate Determination of Project status:-

Helps to track configuration management tool helps to track various deliverables objects so that project manager quickly determines current status of the project.

6) Prevent unauthorized access to the work:-

To prevent a software SCM helps to implement a control change process. It therefore becomes possible to prevent unauthorized change to the work.

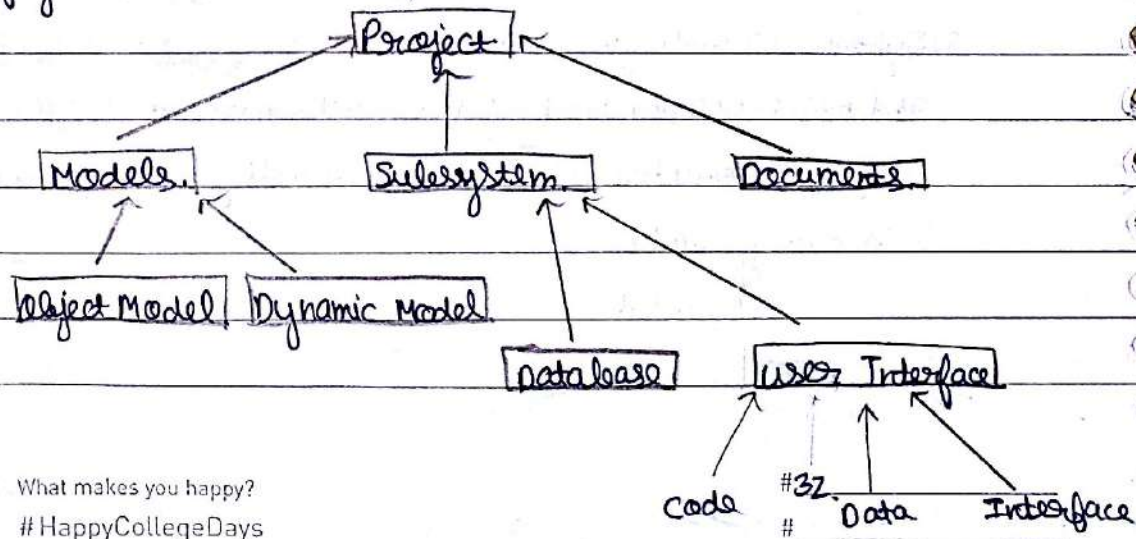
Terminology in SCM:-



① Configuration:-

Configuration of a software is the state of various work products that are under installation control. Work products under configuration control are usually referred as 'configuration item'.

Configuration items (Example):-



Promotion Management → marketing → done at developer's end.
 Release Management → release → done at client's end.

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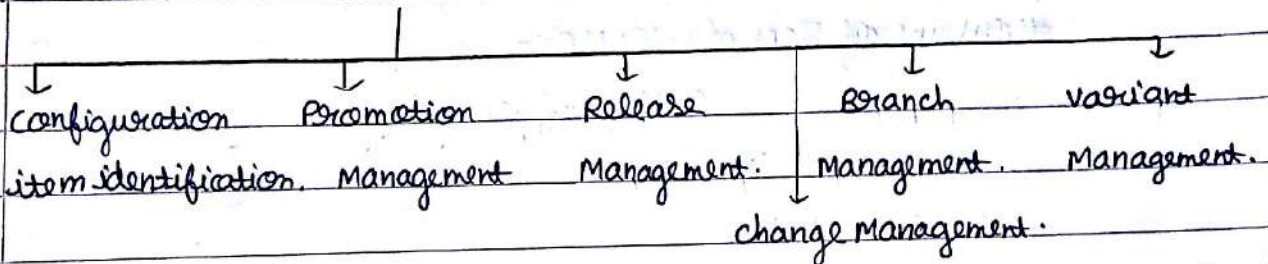
Configuration Items: work products #.

- ② * Version: It is a numerical scheme that helps us to identify a specific configuration at a "certain point of time".
- ③ * Revision: It is a numerical scheme that is used to identify the state of configuration at "any time".
- ④ Baseline: A baseline is a software configurations ^{tool} that has been formally reviewed and agreed upon and serves as a basis or concept for further development.
- ⑤ Variants: Variants are the version that ~~are~~ intended to coexist.

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Activities.



- ① Configuration ~~is~~ Item Identification: modelling of a system as a set of ~~element~~ evolving components.
- ② Promotion management: Promotion Management is the creation of versions for marketing purpose.
- ③ Release Management: It is the creation of version for clients and users.
- ④ Change management: It is the handling, approval and tracking of change requests.
- ⑤ Branch Management: Branch management is the management of concurrent processes working at the same time.
- ⑥ Variant Management: It is the management of variables intended to co-exist.

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Planning → denotes → prototype.
Plan → denotes → document type.

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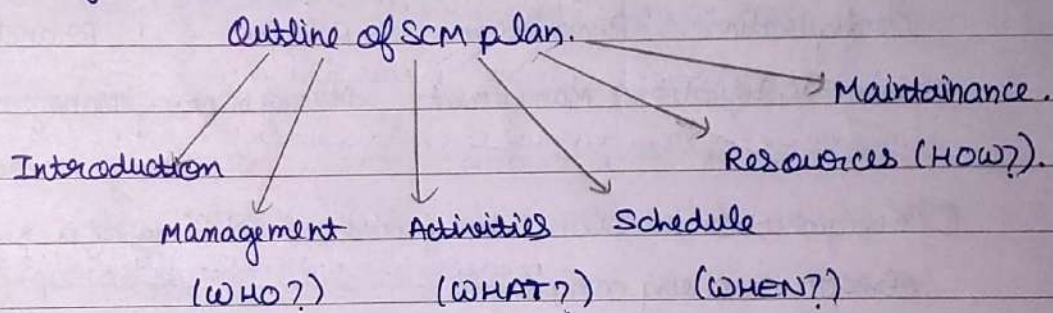
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Software Configuration Management Plan (SCM Plan):-

It defines the type of document to be managed and a document naming scheme.

- ① Define who takes responsibility for configuration management procedure and creation of baseline.
- ② Define policies for change control and version management.
- ③ Describes the tools which should be used to assist configuration management process.
- ④ Define the configuration management database that is used to record configuration information.

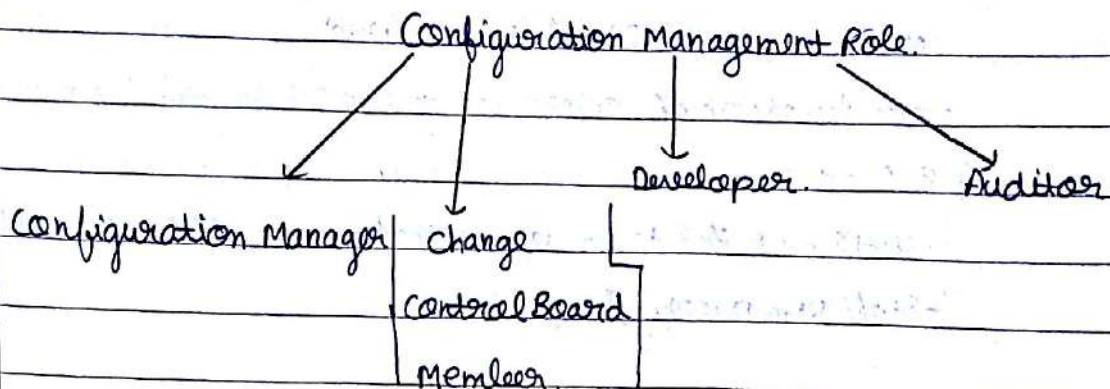
Outline of SCM plan (SCMP):-



- ① Introduction: Describes purpose, scope of application, key terms and preferences.
- ② Management: Identifies the responsibilities and authorities for accomplishing the planned configuration management activities.
- ③ Activities: Identifies the activities to be performed in project.
- ④ Schedule: Establishes the sequences and coordination of SCM activities with project milestones.
- ⑤ Resources: Identifies tools and techniques required for the implementation.
- ⑥ Maintainance: Maintain responsibility that how SCMP will be kept ^(updated) current during life cycle of the project.

* trigger → action.

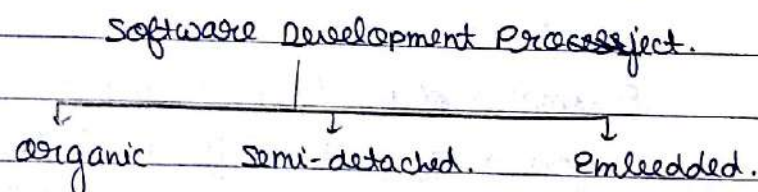
Configuration Management Role.



- * Configuration Manager: Responsible for identifying configuration items. Configuration Manager (CM) can also be responsible for defining the procedures for creating promotions and releases.
- * Change Control Board Members: Responsible for approving or rejecting any change request and in (decision making).
- * Developer: Creates promotions trigger by normal activities of development. The developer checks in changes and resolves conflicts.
- * Auditor: Responsible for selection and evaluation of promotions for release and for insuring the consistency and completeness of this release.

~~# Cocomo~~

- Cocomo Model:- [Cost Constructive Model].



① Organic:-

- less experienced staff is required.
- used for quick, early-designed and rough order of magnitude estimate of software cost.

② Semi-detached:-

- medium size of staff is required.
- combination of experienced and unexperienced team members.

③ Embedded:-

- Highly skilled team is required which works for complex s/w project.
- Staff size is very large.

COCOMO :-

COCOMO Model.

Basic

Intermediate

Detailed.

① Basic COCOMO model:-

$$\text{Effort} = a_1 \times (\text{KLOC})^{a_2} \text{ PM} \quad \left| \begin{array}{l} \text{KLOC} \rightarrow \text{Line of codes in kilos} \\ \text{PM} \rightarrow \text{Person-Month} \rightarrow \text{unit of effort} \end{array} \right.$$

$$T_{\text{dev}} = b_1 \times (\text{Effort})^{b_2} \text{ Months}$$

$a_1, a_2, b_1, b_2 \rightarrow$ coefficients for each category of software products

Estimate of development effort:-

Organic $\rightarrow E = 2.4 \times (\text{KLOC})^{1.05} \text{ PM.}$

Semi-detached $\rightarrow E = 3.0 \times (\text{KLOC})^{1.12} \text{ PM.}$

Embedded $\rightarrow E = 3.6 \times (\text{KLOC})^{1.20} \text{ PM.}$

Estimates of development time:-

Organic $\rightarrow T_{\text{dev}} = 2.5 \times (\text{Effort})^{0.38} \text{ Months.}$

Semi-detached $\rightarrow T_{\text{dev}} = 2.5 \times (\text{Effort})^{0.35} \text{ Months.}$

Embedded $\rightarrow T_{\text{dev}} = 2.5 \times (\text{Effort})^{0.32} \text{ Months.}$

- Q. Assume that the size of an organic type of software product has been estimated to 32000 lines of code. Assume that the average salary of software engineer is ₹ 15000 per month. Determine the effort required to develop the software product and development time also and cost required to develop the product.

Sol: Effort Required = $a \times (KLOC)^{a_2}$ PM.
 $= 2.4 \times (32)^{1.05} \times 15000$ PM.
 $= 1359967.91.33113$ PM.

Now, $T_{dev} = 2.5 \times (91.33113)^{0.38}$ Months.
 $= 13.89 \approx 14$ Months.

$\therefore$ Cost Required = $14 \times 150000 = ₹ 2,10,000$.

② Intermediate COCOMO Model:-

Cost drivers are (Effort Estimator)
 considered.

Project type	a	b
Organic	3.2	1.05
Semi-detached	3.0	1.12
Embedded	2.8	1.20

(Duration Estimator).

Project Type	a	b
Organic	2.5	0.38
Semi-detached	2.5	0.35
Embedded	2.5	0.32

- Q. If a software product is considered as embedded and has been estimated to be 10000 lines of code. Find out its nominal effort.

Sol:
$$\text{Effort} = a_1 \times (\text{kLOC})^{a_2} \text{ PM.}$$

$$= 2.8 \times (10)^{1.20} \text{ PM.}$$

$$= 44.377 \text{ PM.}$$

* Time Size = $\text{Effort} / T_{\text{dev}}$

* Productivity = $\text{Size} / \text{Effort}$

③ Detailed COCOMO Model:- (No numerical is in course for Detailed COCOMO)
It is an extension of the intermediate COCOMO Model that adds effort multiplier for each phase of the project to determine the cost decrease impact on each step.

Result/Conclusion: A major shortcoming of both basic and intermediate COCOMO is - they consider a software product as a single homogeneous entity. However, most large systems are made up of smaller subsystems. The complete COCOMO model considers these difference in characteristics of the subsystems and estimates the efforts and development time as ^{the} sum of estimates for individual subsystems. ~~It~~

COCOMO-II:

The original COCOMO Model has been very successful but it does not apply to newer software development practices.

COCOMO-II targets modern software projects and will continue to evolve over next few years.

COCOMO-II:-

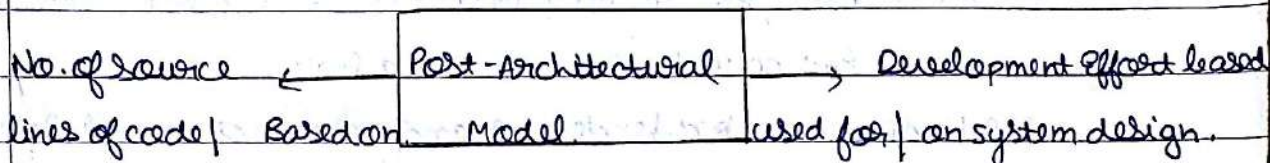
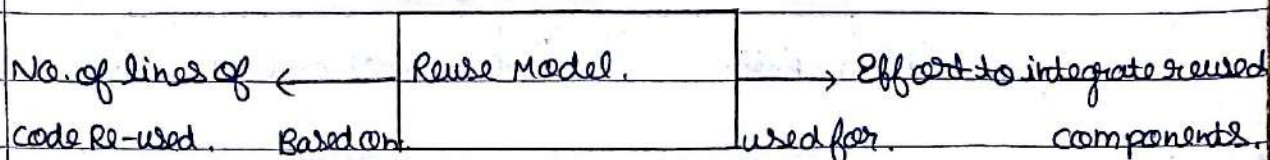
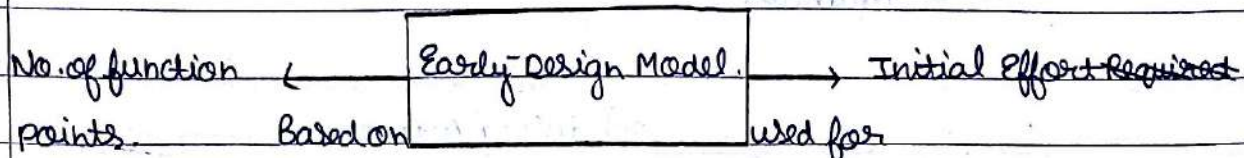
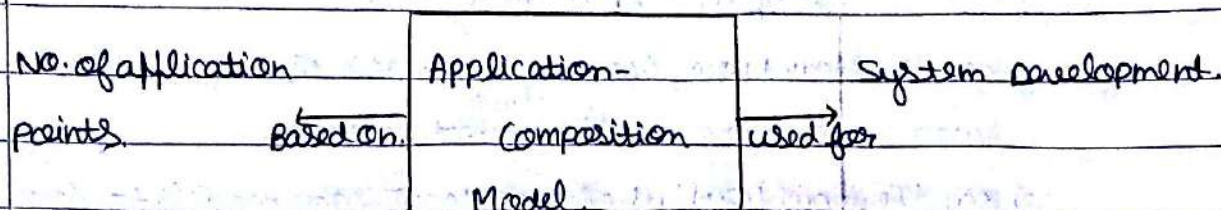
Application Composition Model.

Early-Design Model.

Re-use Model.

Post-Architectural Model.

* Diagrammatic Representation:-



* Application-Composition Model:-

It is used for the earliest phase or spiral cycles.

* Early-Design Model:-

It is used for next phase and involves explorations of architectural alternative or incremental development strategies.

* Reuse Model:-

It is used for calculating reuse cost that indicates the amount of modification of codes reused.

* Post-Architectural Model:-

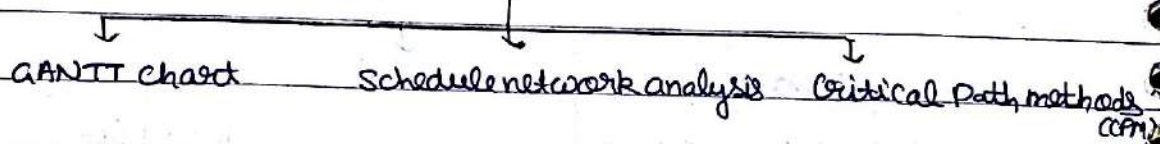
Once the project is ready to develop, it should have a lifecycle architecture which provides accurate information on cost drivers and enables more accurate cost estimates.

* Bar chart: represented vertically; Gantt chart: represented horizontally.

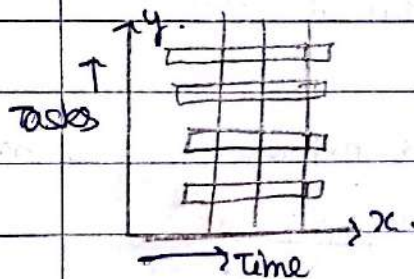
• Strategy Use in COCOMO-II:-

- 1) Preserve the openness of original COCOMO-I.
- 2) Key the structure (contains/consists of) of COCOMO-II to the future software market ^{place} ~~sector~~ sectors.
- 3) Key the input/output of COCOMO-II some models to level of information available.

Scheduling Activities.

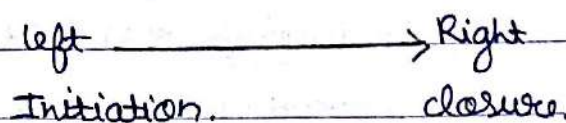


- ① GANTT chart:
 - a) Horizontal bar charts plotted over time i.e. days and weeks. Each activity is shown as a bar.
 - b) Depending on task dependencies and resources availability, these bars may be sequential or parallel.
 - c) Each bar is plotted to start at the earliest possible start date.



② Scheduled Network Analysis:-

- a) Graphical display from left to right across a page of all inter-relationship ^{between} elements of work.



101

Date

01/03/16.

③ Critical Path Method (CPM):-

a) It is the sequential string of activities that takes longest time to complete, recognise any dependency between task in this sequence. CPM is determined by adding the times of all activities of critical path, earliest time that the project can be completed.

• Change Control:-

It is a system approach to manage all changes made to a product or a system.

* Purpose of change control:-

It is to ensure that no unnecessary changes are made, that changes are documented, services are not unnecessarily disrupted and all resources are used efficiently.

* Steps of change control:-

- ① Documenting a change request: When the client requests the change, that request is categorised and recorded along with the informal assessments of the importance and difficulty of implementing it.
- ② Formal Assessment: If the change request is accepted, a development team will be assigned. If the change request is rejected, the fact is documented and communicated to the client.
- ③ Planning: Team responsible for the change create a detailed plan for its design and implementation.
- ④ Designing and Testing: Team designs the program for software changes and test.
- ⑤ Implementation and Review: The team implements the program and
- ⑥ Final Assessment: If client is satisfied, ^{→ change} stakeholder review the change. is implemented satisfactorily and request closed. If client is not satisfied, then project is reassessed and steps may be repeated.

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#Assignment No. 102:-

(Deadline: 08th March, 2016).

- Q1. Explain in detail the monitoring and control process.
- Q2. What are the configuration management activities and also explain need of SCM?
- Q3. Demonstrate the SCM plan and its uses.
- Q4. Give a brief account on COCOMO-II.

• Unit IV: Risk Management:- [Risk Analysis and Management]

• Risk:-

- 1) The expectation of a loss or damage.
- 2) A potential problem (leading to loss) that may or may not occur in the future.
- 3) Risk is a means of measuring the quality of the project.

• Risk Management:-

↳ Analysis + Control.

- 1) It is a set of practices and support tools to identify, analyze and track risk explicitly.
- 2) Risk Management tries to reduce the probability of a risk to occur and the impact caused by it.

• Purpose of Risk Management:-

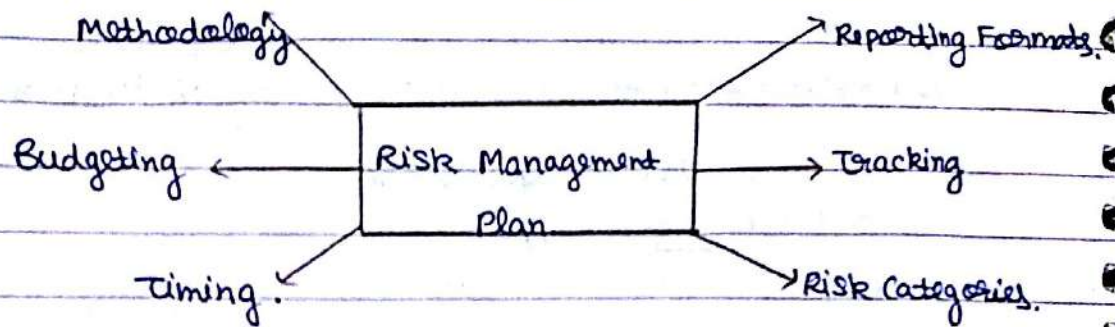
- 1) Anticipate and Identify the risk.
- 2) Minimize the impact/damage/loss.
- 3) ~~Reduce~~ Reduce the probability of occurrence.
- 4) Monitor the risks areas for early detection.
- 5) Ensure Management Awareness.

• Causes of Risk:-

- 1) Lack of Knowledge and Information.
- 2) Lack of Coordination.
- 3) Lack of Time.
- 4) Lack of Resources.
- 5) Lack of ~~Security~~ Security.

• Risk Management Plan:-

A risk management plan describes how the risk management has been done which entails (consists of) identification, qualitative and quantitative analysis, ^{and} monitoring and control.



- ① Methodology: Defines the approaches, tools and data sources used to perform risk management.
- ② Budgeting: Determine a risk management budget for a project and assign resources as required.
- ③ Timing: Establishes how frequently risk management process will be performed throughout the project life cycle.
- ④ Reporting Formats: Describes the content and format of risk response plan. It determines how the results of risk management process are:-

(i) documented
 (ii) analyzed
 (iii) communicated

 } to the project team.
- ⑤ Tracking documents: It is the process of recording all risk capabilities so that information can be used for benefit of current project and to inform future needs.
- ⑥ Risk Categories: Provide a structure for identifying a consistent label of risk detail that contributes to effective and quality of risk identified.

A risk management must include:-

- (i) an introduction describing the content for Risk Management process and general information about Risk Management Plan (RMP)
- (ii) The Risk Management strategy including mitigation plan and risk response plan.
- (iii) The organisation and structure to manage risk.

• Framework of Risk Management:-

- ① Risk Identification.
- ② Risk Analysis.
- ③ Risk Mitigation or risk response.
- ④ Risk monitoring and controlling.

#Explanations:-

① Risk Identification:-

- ↳ Internal.
- ↳ External.

Risk Identification involves determination of the risk that are likely to affect the project and documenting the characteristics of each.

* Risk Identification Address:-

- ↳ Internal
- ↳ External.

a) Internal: Represents issues that the project team can control or influence such as staff assignment and cost estimates.

b) External: represents issues which are beyond the control or ~~or~~ reinforcement of the project team such as market shift or government actions.

* Identifying a risk:-

Source of risk.

- Budget
- Schedule
- Work Breakdown Structure (WBS)
- Objectives
- Project Requirements.

* Participants in risk identification process:-

- (i) Project team.
- (ii) Risk Management team.
- (iii) Customers.
- (iv) Other Project Managers and Stakeholders.

* Categories of Risk:-

- Technical Quality
- Project Management Risk.
- Organizational Risk.
- External Risk.
- Political Risk.

* Tools and techniques used in risk identification:-

- Project charter.
- Schedule estimates.
- Product description.
- Cost estimates.
- Resource plan.
- Assumptions.

② Risk Analysis:-

Risk analysis is about understanding and evaluating the identified risk associated with a project and determining which risk event warrant (guarantee) a response.

A project risk event status is determined from a combination of probability and impact. Risk analysis can be carried using two types of approach:-

a) Qualitative approach.

b) Quantitative approach.

- ① Qualitative approach ^{of risk analysis:} Aims of qualitative risk analysis include:
- (i) Assessing likelihood of identifying risk occurrence.
 - (ii) Assessing impact of risk.

* Definition:-

Risk based on their potential effect on project objectives. This type of analysis determine the importance of addressing specific risk and guiding risk response.

* Qualitative Methods:-

These methods can be used when the level of risk is low and does not warrant (guarantee) the time and resources necessary for making full analysis. It includes:-

- (i) Questionnaire and structured interviews.
- (ii) Evaluation of multi-disciplinary groups.
- (iii) Judgement of specialist and techniques.

- ② Quantitative approach of Risk analysis: The aim of quantitative risk analysis is to analyse numerically the probability of each risk occurring and assessing the consequences on project objectives.

* Quantitative Methods:-

These methods are considered to be those that enable us to assign the values of occurrence to the various risk identified that is calculating the level of risk of the project. It includes:-

- (i) Analysis of likelihood i.e. probability.
- (ii) Analysis of consequences.

③ Risk Mitigation (Risk Response):-

This is the process of finding appropriate responses to the risk. Risk mitigation involves an attempt to reduce the probability or impact of an adverse risk event to an acceptable threshold.

* Strategies of Risk Mitigation:-

- a) Assume/Accept.
- b) Avoid.
- c) Control.
- d) Transfer.
- e) Watch/Monitor.

* Explanation:-

- ① Assume/Accept: Acknowledge the existence of a particular risk and make a deliberate decision to accept it without engaging in special efforts to control it.
- ② Avoid: Adjust program requirements or constraints to eliminate or reduce the risk.
- ③ Control: Implement actions to minimize the actions/impact of the risk.
- ④ Transfer: Reassign organizational accountability, responsibility and authority to another stakeholder who is willing to accept this risk.
- ⑤ Watch/Monitor: Monitor the environment for changes that affects the nature of risk.

④ Risk Monitoring/Controlling:-

This process involves:-

- Tracking identified risk.
- Monitoring residual risk. (Risk which is left in the system).
- Identifying new risk.
- Ensuring the execution of risk plan.
- Evaluating the effectiveness of risk plan in reducing risk.

* Purpose of Risk Monitoring/Controlling:-

- To check if the risk responses have been implemented as planned.
- To check if the risk response actions are as effective as expected.
- To check project assumptions are still valid or not.
- To check if risk trigger has occurred.
- If proper policies and procedures are followed.

* Elements of Monitoring/Controlling:-

- Risk Management Plan (RMP).
- Risk Response Plan (RRP)
- Additional Risk Identification and Analysis.
- Scope changes.
- Project Communication.

Risk Response Plan (RRP): It lists agreed risk responses for specific project and should contain the level of detail for risk response actions.

Project Communication.

↳ Work Results

↳ Reports.

∴ Work Results: Providing information on project performance.

∴ Reports: Documents/Summaries generated to monitor and control risks.

Additional Risk Identification and Analysis: Re-implement the cycle of risk processes.

Audit $\rightarrow$ Examining / Reevaluating.

Scope changes: Require new risk analysis and response plan.

* Methods of Monitoring / Controlling:-

- a) Project Response Audits.
- b) Periodic project risk reviews.
- c) Earned value analysis.
- d) Technical performance measurement.
- e) Additional risk response planning.

* Explanation:-

@ Project Response Audits: (i) Risk auditors examine and document the effectiveness of risk response in avoiding, transferring or mitigating risk occurrence.

(ii) These audits also evaluate the effectiveness of risk owner.

(iii) Risk audits are carried during the project life cycle to control risk.

@ Periodic Project Risk Review: (i) Risk rating and prioritization may change during the project lifecycles.

(ii) Reviews should be scheduled regularly and appear as an agenda item at all team meetings.

@ Earned value analysis: (i) It is used for monitoring overall project performance against a baseline plan.

(ii) Results may indicate a potential deviation of project at completion from: $\therefore$ cost $\xrightarrow{CPI}$ $\therefore$ schedule, (SPI).

(iii) Earned Value Analysis (EVA) is the standard method of measuring the project progress at any given point of time, forecasting its completion date and final cost and analyzing variances in schedule and budget as the project proceeds.

It compares the planned account of work with what has

actually been completed, to determine the cost, schedule and work accomplished are progressing in accordance with plan.

④ Technical performance measurement: (i) It compares technical accomplishments during project execution to project plan schedule of technical achievements.

(ii) Any deviation from the project plan schedule can indicate a risk to achieve the project scope.

⑤ Additional risk response planning: (i) If a risk has a greater impact than anticipated impact on objectives, then planned response may not be adequate / sufficient.

(ii) If there is an emerging risk not previously anticipated in the risk response plan, then plan response may not be adequate.

• software metrics → already given in assignment no. 03.

#unit IV: Managing people:-

• Managing people working in software environment:-

Management activities.

- problem solving (using available people)
- Motivation (people working on project)
- Planning (what people are going to do?)
- Controlling (people activities)
- Estimating (how fast people will work)...
- Organising (the way in which people work).

• Main concerns of Managing people :-

- Staff selection
- Staff development
- Well-being staff during the course of project.

• Understanding Behaviour:-

understanding behaviour means to identify how to handle the people (handling people). Approaches for understanding behaviour are as follows:-

- Positivist approach.
- Interpretivist approach.

- ① Positivist approach: (i) This approach is based on development of system.
(ii) Tends to be objective and empirical.
(iii) Seeks cause for behaviour.
(iv) Conducts studies that can be generalised to larger population.
- ② Interpretivist approach: (i) This approach defines how software

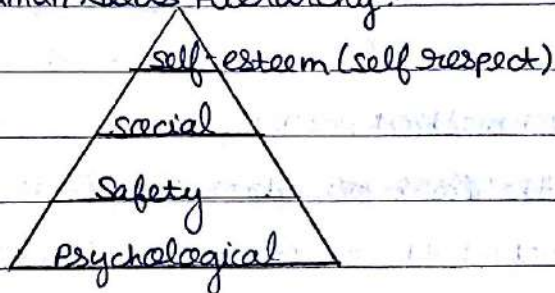
Engineer is defined.

(ii) who will customize and install the software package.

(iii) This approach is more qualitative and based on smaller samples.

• Motivation:-

* Human Needs Hierarchy:-



Motivation.

- Task-Oriented
- Self-oriented
- Interaction-oriented

An important role of a manager is to motivate the people working on a project.

Motivation has three different types:-

- ① Basic Needs. (food, job security).
- ② Personal Needs. (self-respect)
- ③ Social Needs. (accepted as a part of group).

* Human Needs

Motivation should also take into account different personality types:-

- ① Task-Oriented.
- ② Self-oriented.
- ③ Interaction-Oriented

* Explanation:-

- ① Task-oriented: Motivation for doing the work is the work itself.

- ② Self-oriented: Motivation to a person to expose his hidden talents/skills.
- ③ Interaction-oriented: Principle motivation in the presence and actions of co-workers.

• Organisational Behaviour:-

Three basic objectives:-

- ① To select the best men for the job.
- ② Instruct them in best method.
- ③ To give incentives in the form of higher wages to the best workers.

The condition under which the staff works also affects productivity.

Two theories were given on organisational behaviour:-

↳ Theory X

↳ Theory Y.

* Theory X:-

- a) An average human has natural tendency of disliking the work.
- b) People tends to avoid responsibility.
- c) There is always a need of direction and control.

* Theory Y:-

- a) Work is as natural as rest or play.
- b) External control is not only the way of bringing efforts directed towards an organisational end.
- c) Commitment to objectives is a function of reward associated with their achievements.
- d) An average human can learn to accept and further seeks responsibility.

• Aldham's and Hackman's Job characteristics Model:-

The Job characteristics Model defined by Aldham~~s~~ and Hackman is based on the idea that the task is itself key to employ motivation. It is one of the only approach that focuses on looking at the job design through person-fit theory.

* Job characteristics:-

- ① Skill Variety
- ② Task Identity
- ③ Task Significance.
- ④ Autonomy
- ⑤ Feedback.

- ① Skill Variety: The degree to which a job requires various activities, requiring the worker to develop a variety of skills and talents. Job holders can experience more meaningfulness in jobs that require several different skills and abilities.
- ② Task Identity: The degree to which job requires the job holders to identify and complete the work.
- ③ Task Significance: The degree to which the job impacts other people's lives. The influence can be either in immediate organisation or in external environment.
- ④ Autonomy: The degree to which job provides employee with significant freedom, independence and discretion to plan out the work and determine the procedures in the job.
- ⑤ Feedback: The degree to which workers has knowledge of results.

• Managing Groups:-

* Factors Affecting Group Working:-

- ① Group Composition.
- ② Group Cohesiveness.
- ③ Group Communication.
- ④ Group Organisation.

* Managing Groups:-

- a) ~~#~~ Most of the software engineering is a group activity.
- b) Group interaction is a key determinant of group performance.

① Group Composition: a) Group consists of members who share different works.

b) An effective group has a balance of all types.

c) Interaction oriented people are very important as they can detect and diffuse tensions that arise.

② Group Cohesiveness: a) In cohesive group, members consider the group to be more important than any individual in it.

Advantage of cohesive group:-

(i) Group quality standards can be developed.

(ii) Team members learn from each other.

(iii) Group members work closely together so inhibitions caused by ignorance are reduced. / inherited from the wrong doings of others.

③ ~~Good~~ Group Communication: a) Good communication is essential for group working.

b) Information must be exchanged on status of work.

c) Good communications also strengthens group unity as it promotes understanding.

- ④ Group Organization: a) Small software engineering group are usually organised informally without a rigid structure.
b) For large projects, there may be hierarchical structure where different groups are responsible for different sub-projects.

• Organisational Structures:-

Organisations are the foundations of software project management that rests on / depends on project management skills, software engineering process and quality system and measurement.

Organisations can also be defined as a group of people who must coordinate their activities in order to meet the organisational objectives.

It has three parts:-

- ↳ ① Informal and Formal Structure
- ↳ ② Departmentalization.
- ↳ ③ Chief Programmer Team.

• Stress:-

already given in assignment

• Decision making process:-

- ① It is another task of effective group.
- ② Amount of risk and uncertainty involved in development process can affect decision making process.
 - ↳ structured.
 - ↳ unstructured.

- a) structured: These are generally simple where rules are implied in a straight-forward way.
- b) unstructured: These are more complex and requires a greater degree of creativity.

• Health and safety:-

- ① There are two important factors that are prominent in constitution of other engineering heavy projects are health and safety.
- ② A project manager must be aware of safety policy document that applies to environment in which product is developed.
- ③ Safety policy document must be maintained by manager to meet safety objectives such as reliability of completing application or overall cost of project.

• Different levels of safety issue:-

- ① Top management must abide by safety policy.
- ② Delegation of responsibilities for safety must be clear.
- ③ Job description should include duties related to safety.
- ④ Adequate budget for safety cost.
- ⑤ Safety procedures must be known to employee and necessary health must be given whenever needed.